

RESEARCH ARTICLE

DEM Investigation on the Collecting Characteristics and Performance Evaluation of Lunar Soil Within a Hollow Drilling Tool

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ABSTRACT

Lunar soil collection is pivotal for advancing extraterrestrial exploration, enabling in-situ mineral resource utilization, sustainable lunar base construction, and scientific investigations into lunar geological history. In this study, the collection behavior of lunar soil within a hollow drilling tool was investigated using the Discrete Element Method (DEM) based on the Hertz-Mindlin contact model. The effects of key parameters of drilling tool in direct contact with the sample, including inner diameter of the drill pipe (D), friction coefficient (μ), and height of the complement interior the drill bit (H) on macroscopic properties (sample mass, coring ratio, uniformity) and microscopic properties (pore size, disturbance) were systematically analyzed. Results demonstrated that increasing D or decreasing μ enhances sample mass and disturbance rate, while reducing H increases sample mass but reduces disturbance rate. Collection law analysis revealed that the sample can be divided into three regions: the new sample region, the radial movement region, and the flow region. Mechanism analysis uncovered that the energy loss caused by collision and friction changes the average velocity of particles relative to the drilling tool, and the horizontal force arch formed between particles hinders the particle motion. The response surface method (RSM) was employed to predict sample mass and disturbance rate, demonstrating high accuracy ($R^2 > 0.95$). The findings provide valuable insights for optimizing lunar soil sampling tools and offer practical guidance to balance efficiency and sample integrity in extraterrestrial missions.

1 | Introduction

In recent years, the moon has attracted increasingly attention from scientists and engineers owing to its unique spatial location, abundant minerals and broad prospects for scientific exploration [1–3]. A number of pioneering technologies have been proposed for establishment of lunar bases by utilizing lunar soil, including additive manufacturing [4, 5] and extraterrestrial artificial light synthesis [6]. Therefore, the collection and analysis of lunar soil is particularly important for the utilization of lunar in-situ resources and the development of related instrumentation. The Spiral drilling tool, featured with excellent adaptability and

effectiveness in different geological conditions, has been applied in various successful extraterrestrial exploration missions to date, including the US Apollo Program [7], the former Soviet Union's Luna Program [8], and the Chinese Lunar Exploration Program [2]. In contrast to soft-bag drilling tool which is limited to single fixed-position sampling, while the hollow drilling tool is capable of being transported by the lunar rover for multiple removable sampling missions. The drilling tool parameters exert a direct influence on the metrics pertaining to the sampling missions, such as sample mass, porosity, and disturbance. Consequently, it is crucial to understand the drilling tool parameters with a view to investigating the kinetic and mechanical characteristics

of lunar soil and improving the efficiency of the sampling missions.

Thus far, three methods have been applied to improving the efficiency of the sampling missions: empirical research, theoretical analysis and numerical simulation. Physically, the scarcity of lunar soil has led to the creation of lunar soil simulants that resemble real lunar soil in terms of density, particle size distribution, and mechanical characteristics, such as JLU [9] and JSC-1 [10], have been tested for their suitability for drilling. Tang et al. employed an ultrasonic sensor to monitor sample height and coring ratio online, thereby demonstrating the effect of the penetration per revolution index on the kinetic characteristics [11]. Pang et al. demonstrated that the porosity of the lunar soil has a significant effect on the loads of drilling tool and investigated the relationship between the internal stress of lunar soil and the bulk density [12]. Zhang et al. conducted a series of semi-time and full-time cutting tests at varying cutting speeds, cutting depths, soil densities, and tip angles to gain a deeper understanding of the interaction between the time and the soil [13]. Moreover, Xu et al. constructed a load predictor and a temperature predictor based on the real-time extracted drilling state parameters, and successfully predicted the penetration force, rotational torque, and temperature on the drilling tool accurately during the next second [14, 15]. Previous work indicates that although empirical research can monitor macroscopic targets like sampling height and load, it is still a challenge to observe the kinetic characteristics and disturbance of the lunar soil. Furthermore, the financial investment required to reproduce the lunar environment like low gravity in laboratory is significant. Fortunately, these issues can be effectively addressed through theoretical and numerical models.

The focus of most theoretical analysis has been primarily oriented toward the impact of various factors on the load of drilling tool, such as filling rate of external rotating thread, drilling operating parameters, and drilling structural parameters through the application of the rotary cutting theory of the drill bit and the soil transport theory of the external rotating thread. Based on the principle of equal volume, Tian et al. proposed a theoretical model on power consumption and soil conveying volume, and then established a multi-objective function to optimize the structural parameters of external rotating thread [16]. Furthermore, based on the Mohr-Coulomb soil mechanics, Zhang et al. developed a theoretical model for predicting the penetration force and rotational torque of drilling tool [17]. Additionally, based on the spiral transport theory, Quan et al. incorporated the filling rate of external rotating thread into the theoretical load model, thereby facilitating an investigation into the penetration force and rotational torque exerted on the drilling tool under conditions of non-choking and choking [18]. Although theoretical analysis is useful for predicting macroscopic targets, it unfortunately falls short in visualizing the motion of particles during the sampling process. Therefore, theoretical analysis is less effective in elucidating detailed scientific issues such as the kinetic characteristics and disturbance of lunar soil.

Regarding numerical simulation, Discrete Element Method (DEM) is more appropriate than Finite Element Method (FEM) for analyzing discontinuous medium like lunar soil [19–22]. In this aspect, Gao et al. demonstrated the significance of preserving stratification information on in situ lunar soil and investigated

the impact of friction coefficient on the porosity and stress distribution of soil during direct push sampling [23]. Meanwhile, Wang et al. highlighted the crucial role of adjusting the structure of drilling tool and the contact between the drill tool and the soil in optimizing the drilling process [24]. Liu et al. were devoted to establishing an alterable constitutive law to improve the accuracy of the lunar soil model, and suggested that the presence of complement formed interior the drilling bit contributes to the increase in sample mass [25, 26]. Additionally, Chen et al. [27] and Zhang et al. [28] utilized the knowledge of bionics to reduce the specific energy consumption and improve the sample mass. These studies validate the effective application of DEM to lunar soil, particularly in the sampling process [29]. Despite the variety of research perspectives and methodologies employed in the field of drilling, most studies have concentrated on the macroscopic properties, such as the sample mass, perturbation, and temperature field. This has resulted in a paucity of analysis of the effect of drilling tool parameters on the microscopic properties of samples, the collection law, and the mechanism.

In this article, the sampling process of lunar soil constructed in a hollow drilling tool based on the Hertz-Mindlin model is numerically reproduced by EDEM 2024. Firstly, the accuracy of the calibrated simulation parameters and the established numerical model is validated by comparing particle physical properties and sampling results with previous work. Subsequently, the impact of the key parameters of drilling tool in direct contact with the soil (the inner diameter of the drill pipe D , the friction coefficient μ and height of the complement interior the drill bit H) on the macroscopic properties (sample mass, coring ratio, and uniformity, etc.) and microscopic properties (pore size and disturbance) are discussed. Then the collection law is analyzed by observing the positional changes of the particles during sampling, and the corresponding mechanism is explored in combination with the velocity and normal force of the particles. Finally, prediction models on sample mass and disturbance rate are respectively developed, providing in-depth and comprehensive insights on the sampling process of lunar soil.

2 | Numerical Method and Simulation Conditions

2.1 | Governing Equations

The Hertz-Mindlin no-slip model represents a classical discrete element contact model, frequently employed for the simulation of lunar soil [30]. In this model, the motion of particles is driven by the normal and tangential contact forces concurrently based on the Newton's Second Law. Specifically, the normal force followed the classical Hertz's theory, while the tangential force is based on the Mindlin and Deresiewicz simplifications. The normal contact force F_n and normal damping force F_n^d are respectively expressed as:

$$F_n = \frac{4}{3}E^*\sqrt{R^*\delta_n^{\frac{3}{2}}} \quad (1)$$

$$F_n^d = -2\sqrt{\frac{5}{6}}\beta\sqrt{S_n m^*} \cdot \mathbf{v}_n^{rel} \quad (2)$$

where E^* and R^* are the equivalent Young's modulus and the equivalent radius respectively; δ_n is the normal overlap; β is

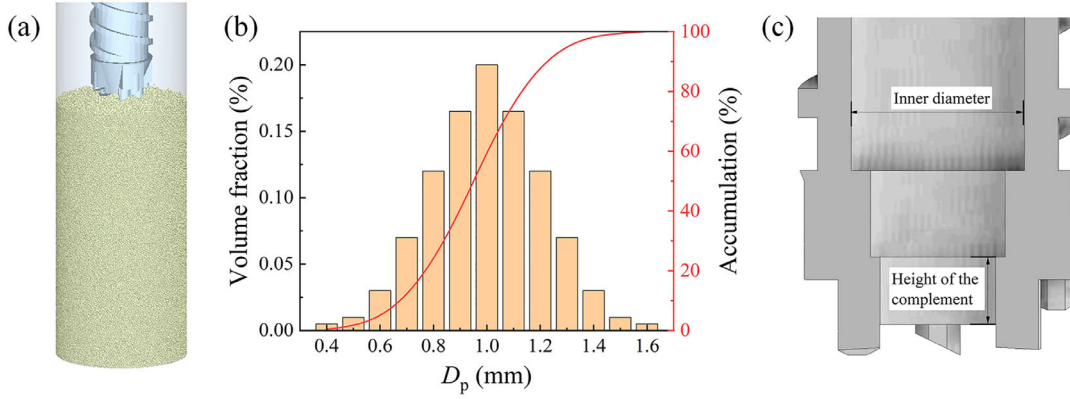


FIGURE 1 | (a) Initial state of the drilling model; (b) size distribution of particles used in the simulations; and (c) geometry of the drilling tool.

the damping coefficient; S_n is the normal stiffness; m^* is the equivalent mass and $\mathbf{v}_n^{rel}$ is the normal component of the relative velocity.

The tangential contact force F_t and damping force F_t^d are respectively given by:

$$F_t = -S_t \delta_t \quad (3)$$

$$F_t^d = -2\sqrt{\frac{5}{6}}\beta\sqrt{S_t m^*} \cdot \mathbf{v}_n^{rel} \quad (4)$$

where S_t and δ_t are the tangential stiffness and tangential overlap; $\mathbf{v}_n^{rel}$ is the relative tangential velocity. The detailed force model can be found in our previous studies [31, 32].

The rigid boundary is simplified from a surface to a plane by triangular mesh division, and the interaction of particles with triangular mesh is modelled according to Hertzian contact theory [33].

2.2 | Simulation Conditions

Figure 1a illustrates the initial state of the drilling model, consisting of lunar soil particles, a drill pipe with an external helix, a drill bit with an interior complement, and a cylindrical container. Based on the similarity principle and coarse-graining technique that have been applied in DEM simulation of lunar soil [34, 35], about 390,000 spherical particles following a Gaussian size distribution in the range of 0.4 ~ 1.6 mm were simulated in the current work (Figure 1b). First, the macroscopic parameters are directly derived from widely recognized real lunar soil parameters, such as particle size distribution [36], particle density [37], and porosity [38]. Second, the parameters (including Young's modulus, Poisson's ratio, restitution coefficient, and various friction coefficients) are determined through systematic calibration against the experimentally validated angle of repose of lunar soil simulants [9] in Section 2.3.1. The geometry of the drilling tool is shown in Figure 1c, and the physical and structural parameters are consistent with the work of Liu et al. [25]. Detailed parameters of the particles and drilling tool are listed in Table 1. To ensure the reliability of the simulation results, the DEM time

TABLE 1 | Particle and tool parameters utilized in simulations.

Parameters	Values
Diameter of particles, D_p (mm)	0.4–1.6 [36]
Density of particles, ρ_p (kg/m ³)	2900 [37]
Initial porosity of particles, (-)	0.465 [38]
Young's modulus of particles, E_p (MPa)	50 [9]
Poisson's ratio of lunar soil simulant, ν_p (-)	0.25 [9]
Density of steel, ρ_s (kg/m ³)	7850 [9]
Young's modulus of steel, E_s (GPa)	184.9 [9]
Poisson's ratio of steel, ν_s (-)	0.3 [9]
Height of the bit, H_b (mm)	16 [25]
Inner diameter of the bit, R_{ib} (mm)	14 [25]
External diameter of the stem, R_{es} (mm)	25 [25]
Feeding speed, n (mm/min)	120 [25]
Rotational speed, v_p (rpm)	120 [25]
Gravity, g (m/s ²)	1.6
Diameter of boundary, R_b (mm)	60
Restitution coefficient of particles, -	0.507
Static friction coefficient of particles, -	0.404
Rolling friction coefficient of particles, -	0.069
Restitution coefficient between particles and steel, -	0.695
Static friction coefficient between particles and steel, μ (-)	0.618
Rolling friction coefficient between particles and steel, -	0.051

step is set to 30% of the Rayleigh time step as given by [39, 40]:

$$T_R = \frac{\pi D_p}{2} \left(\frac{\rho_p}{G} \right)^{\frac{1}{2}} (0.1631\nu_p + 0.8766)^{-1} \quad (5)$$

where D_p is the particle diameter; ρ_p is the particle density; G is the shear modulus, and ν_p is the Poisson's ratio.

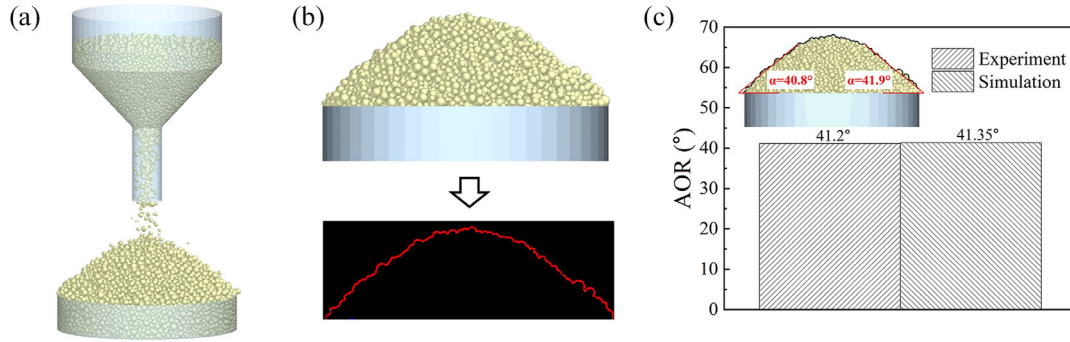


FIGURE 2 | (a) The geometric model of virtual funnel test; (b) process of numerical AOR measurement; and (c) comparison of the AOR between physical and numerical experiments.

2.3 | Model Validation

2.3.1 | AOR Test

The AOR represents the angle formed by the free slope of the powder pile with the horizontal plane, which is a macroscopic manifestation determined by the microscopic parameters between particles and between the particles and the geometry. The funnel test constitutes the standard methodology for measuring AOR, and the experimental setup is illustrated in Figure 2a. The microscopic parameters are calibrated by fitting AOR values derived from numerical and experimental funnel tests. To accurately measure the AOR, a home-made code is used to identify and fit the contour line of the stable powder pile as shown in Figure 2b. The calibration process is conducted using the Box-Behnken method, which has been successfully utilized in our research [41]. Following the construction of a regression model for the AOR, the model is optimized with the actual AOR as the target within the range of the test factors. Figure 2c shows that the optimized AOR in the numerical experiment is 41.35°, which agrees well with others' results [9]. The detailed numerical test results of AOR are presented in Table A1 of Appendix A1.

2.3.2 | Triaxial Compression Test

Triaxial compression test is a method of measuring the stress-strain characteristics of soil by loading the soil radially while maintaining lateral pressure. Scott [38] performed triaxial compression tests on real lunar soil and obtained stress-strain curves for the loose and dense sample. Figure 3a shows the simulated triaxial compression test process. In this study, 20 mm × 20 mm × 42 mm specimen is first constructed, and then six-sided circumferential pressure P is applied until the particles remain stable. The axial planes are then compressed inwards at a strain rate v_a of 0.3%/s until the axial strain reaches 20%. Finally, the curves of axial stress difference versus axial strain are shown in Figure 3b. One can find that the simulation and experiment results are much comparable, indicating that the parameters of the lunar soil particles can realistically reflect the physics-mechanical properties during both the initial drilling stage and the later compression stage.

2.3.3 | Equivalent Boundary

A container that is excessively narrow or shallow may alter particle flow behavior, whereas an overly large container reduces computational efficiency due to the increased number of particles. Therefore, a proper sized container is required to ensure that the particles located on the boundary maintain low velocities and forces, thereby ensuring accurate sampling model. The velocity field of the particles at the final moment is illustrated in Figure 3c, and Figure 3d shows the axial stress along the Y direction. As can be seen that the particles move vigorously around the drilling tool, while both velocity and stress decay with the distance away from the tool. Eventually, particles are nearly stationary when the stress is reduced to 0.45 MPa at the boundary. In accordance with Newton's First Law, it can be established that there is no significant change in the position of particles at the boundary, and the boundary is deemed reasonable. On this basis, a moving surface is set on the inner wall of the drill pipe to simulate the process of constantly inward folding of the soft bag as shown in Figure 3e. The sample is defined as the particles that enter the drill pipe, excluding those particles that are located within the drill bit. The sampling height is 50.32 mm, and the error with the result of literature [25] is 3.4%. Therefore, it can be concluded that the drilling model used is accurate.

3 | Results and Discussion

3.1 | Effects of the Inner Diameter of the Drill Pipe

3.1.1 | Collection Performance Analysis

The sample mass m_s is a critical parameter in assessing the suitability of samples to meet mission requirements. Figure 4 shows the variation curves of m_s and instant collection rate τ_m under the effect of the inner diameter of the drill pipe D . Initially, the particles are compressed until the drill bit is fully inserted into the particles, so the sample delivered by the bit to the drill pipe is identical across all cases. After that, the m_s increases with time until it approaches saturation (i.e., $\tau_m < 0.2$ g/s [29]) as shown in Figure 4a. An increasing trend in the m_s is observed with an increasing D , reaching a maximum value of 11.24 g when D is 22 mm. Besides, the τ_m exhibits a rapid increase at the same rate after the particles entering the drill pipe, reaching a maximum

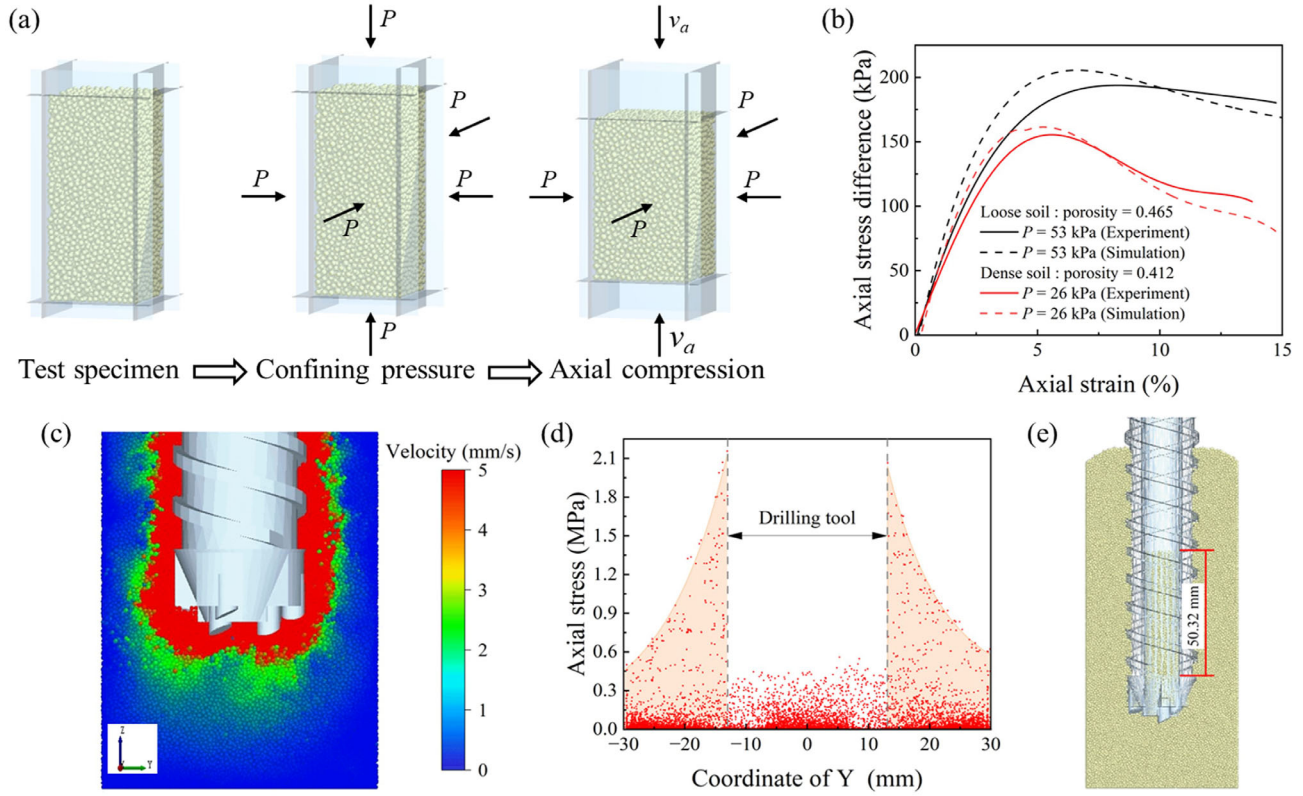


FIGURE 3 | (a) Procedure of triaxial compression test; (b) comparisons of the stress-strain curves between experiment and simulation; velocity field (c) and axial stress (d) of particles at final moment; and (e) sampling height of simulation with soft bag.

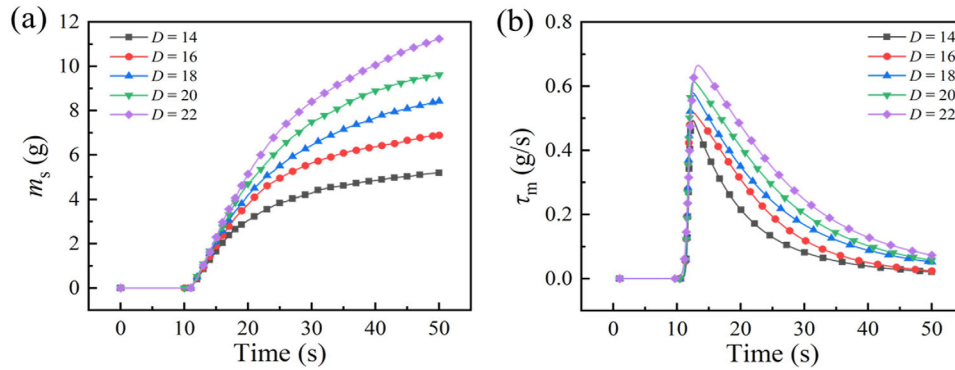


FIGURE 4 | Evolutions of m_s (a) and τ_m (b) at different inner diameter of the drill pipe D .

value before decreasing slowly to a steady state, as illustrated in Figure 4b. This is because in the cases where the particles are not in contact with the inner wall of the drill pipe, the resistance is minimal, allowing free movement. Concurrently, the larger the D , the later the particles contact the inner wall of the drill pipe, which is manifested by a later and larger peak of τ_m . Subsequently, the motion of the particles is gradually resisted, resulting in the decrease of τ_m , but the larger τ_m is still maintained when D is larger because of the motion space. Thus, increasing m_s can be achieved by increasing D .

Meanwhile, the sample height H_s and the coring rate K_c allow for the accurate reflection of the status of the samples in real time (e.g., the presence of jammed drills [42]). Figure 5a shows the

increase of H_s at the same drilling depth as the D decreases, which is consistent with previous study [25]. Moreover, the instant increase rate τ_h demonstrates a non-monotonic pattern over time as shown in Figure 5b, and its evolution can be divided into three stages. In Stage I (arch-forming stage), an arch of the same height is formed when the particles enter the drill pipe, so that similar peaks in the τ_h are observed in different D . In Stage II (empty space filling stage), the particles entering the drill pipe primarily collapse around to fill the empty space created by the difference between the inner diameter of the pipe and the bit. The collapse behavior is significant at larger inner diameter, which leads to a sharp decrease in τ_h . Subsequently, the larger the D , the later the resistance of the drill pipe to the movement of the particles is formed, resulting in the presence of a period of increase in the τ_h .

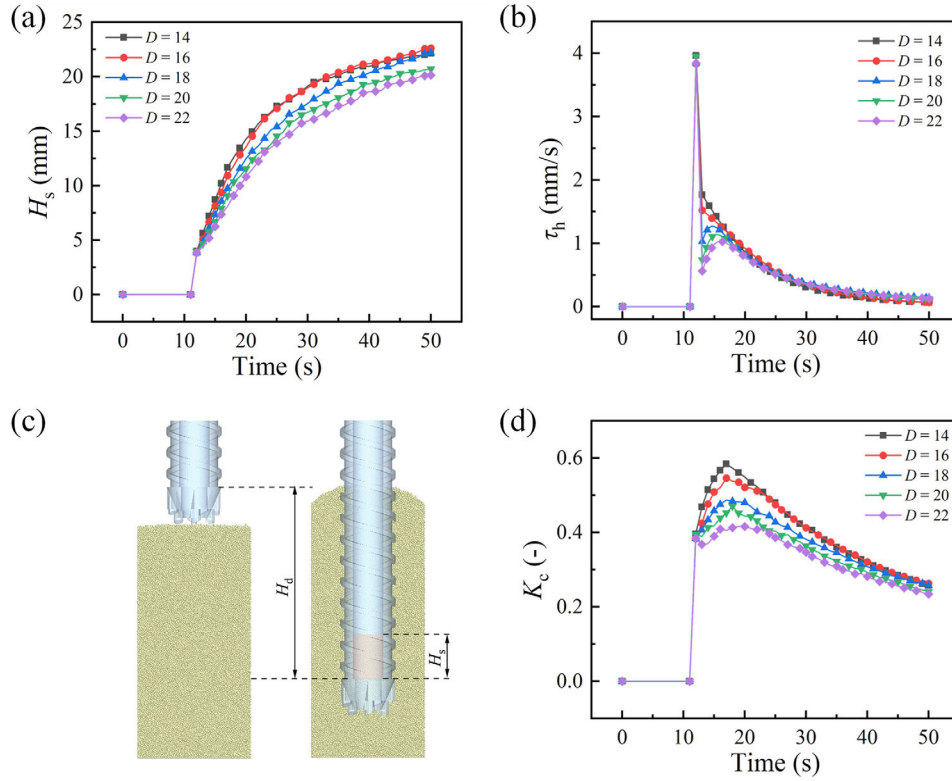


FIGURE 5 | Evolutions of sample height H_s (a) and instant increase rate τ_h (b) at different D ; (c) core ratio characterization; and (d) evolution of coring rate K_c at different D .

If the D is equal to or slightly larger than the inside diameter of the drill bit, the space that can be filled is minimal, and the resistance acts quickly, thus resulting in no rising phase in the τ_h . In Stage III (height growing stage), the resistance of the particles to enter the drill pipe becomes continuously stronger, causing τ_h to start decreasing gradually until it reaches zero. Moreover, the restricted motion space constrains the radial movement of particles, and the height increases significantly.

The coring rate, K_c , is defined as the ratio of the sample height H_s to the displacement of the drill pipe H_d , as illustrated in Figure 5c. A high K_c has been found to reduce the frequency of sampling missions to minimize energy loss and equipment deterioration. As demonstrated in Figure 5d, the K_c first increases and then decreases [42]. This phenomenon can be attributed that at initial stage of drilling, during which H_d is relatively small whereas H_s increases rapidly. This reduces the difference between H_s and H_d , thereby enhancing the K_c . Over time, as τ_h decreases and H_d grows larger, the gap between H_s and H_d widens, leading to a decline in the K_c . Consequently, the utilization of drill pipe with a small D has been proved to facilitate a greater H_s .

In addition to m_s and H_s , the porosity and homogeneity of the sample are also used to quantitatively characterize the quality of the sample. Indeed, homogeneous and dense samples are less likely to undergo changes in their pore size during transport and are more amenable to preservation. To provide a clear description of the distribution of particles, the sample is divided into cylinders with a height of 4 mm, as illustrated in Figure 6a. The relative standard deviation (σ), which is a measure of the uniformity of the particle distribution [43], is calculated by:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (N_i - N_{av})^2} \quad (6)$$

where N is the number of layers, N_i is the porosity of layer i , and N_{av} is the average porosity. Figure 6b shows that the porosity and σ decrease as D increases. This means that the particles have more space to move in a drill pipe with a larger D and by accelerating the rearrangement of the particles, the pores are disrupted, leading to a reduction in porosity. At the same time, the movement of the particles leads to an increase in the homogeneity of the sample. Notably, the porosities are all less than the original lunar soil, suggesting that the samples are squeezed.

The pore size of the sample formed by the particles is also characterized quantitatively using a technique known as Radical Tessellation (RT) [44]. In this method, the whole sample is divided into a set of non-overlapping convex Radical Polyhedrons (RPs), each RP containing one particle. Thus, the volume fraction δ of pores around each particle can be calculated by:

$$\delta = \frac{V_{rp} - V_p}{V_{rp}} \quad (7)$$

where: V_{rp} is the volume of each RP; V_p is the volume of the particle enclosed in RP. A larger δ means a larger volume percentage of pore. Figure 6c shows the visualization of the RT network. The effect of D on the δ and porosity of the first four layers is shown in Figure 6d. Firstly, in Layer 1, particles exhibit low δ within the diameter range of the drill bit, forming a compression zone. Subsequently, the particles are in a turbulent

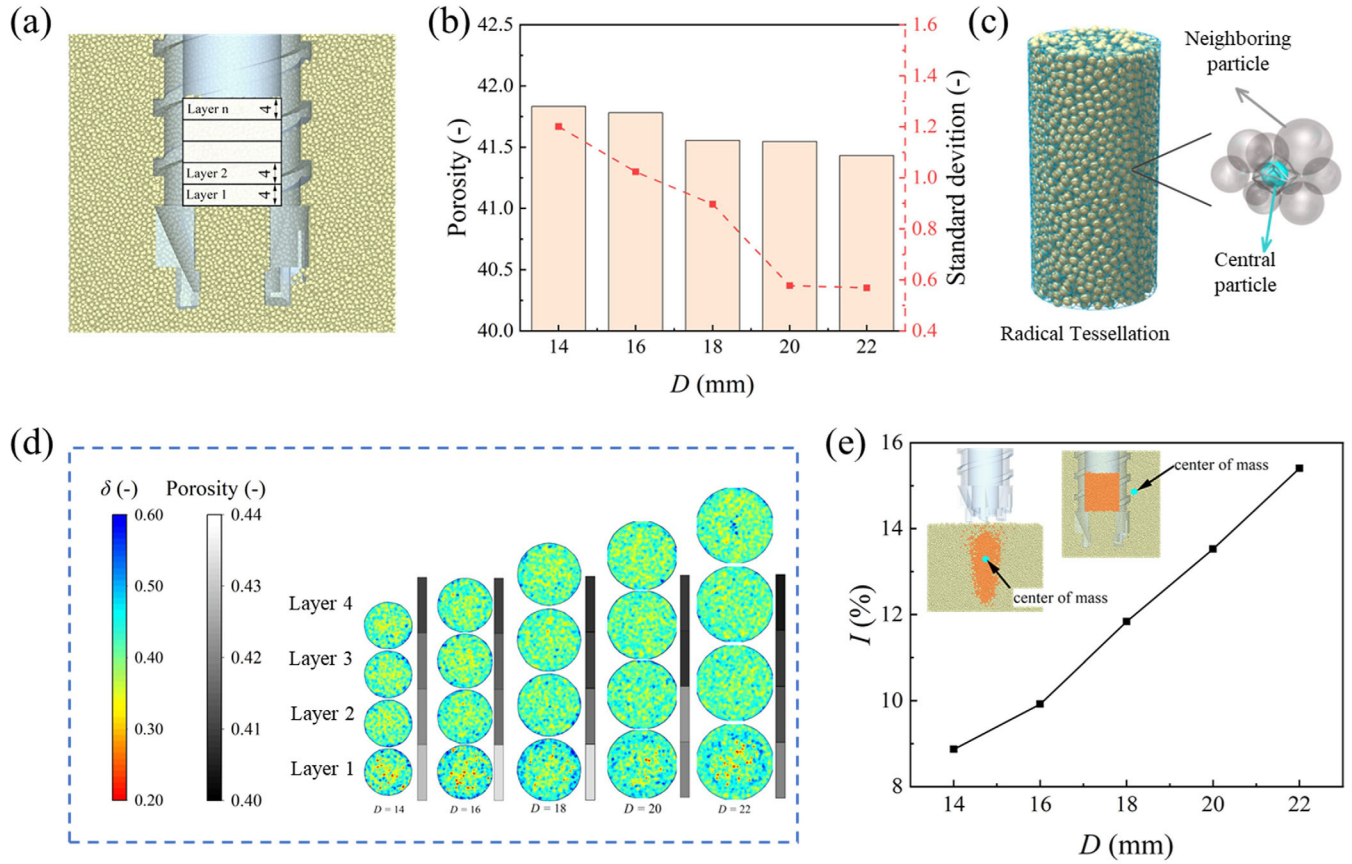


FIGURE 6 | (a) Sample stratification; (b) evolution of porosity and homogeneity at different D ; (c) Radical Tessellation; evolutions of δ and porosity of different layers (d) and I (e) at different D .

state caused by the high velocity and force and must be rearranged before they contact the inner wall of the drill pipe, forming a rearrangement zone where the δ is high. When the D is about equal to the inner diameter of the drill bit, the rearranged zone is close to the inner wall of the drill pipe, making the δ at the boundary high. Notably, a larger D provides more motion space, and the sufficient movement of particles leads to the lowest porosity of the first layer compared with other D . As the number of layers increases, the particles gradually are stabilized and the pores are gradually filled, resulting in the densification of the sample. This is the reason for the phenomenon that the porosity of the first layer is larger than that of other layers.

Disturbance is caused by the relative displacement of particles during sampling. With consideration of the height of the sample, the mean value of the displacement of the particles relative to the center of mass of the sample before and after sampling is called the disturbance rate I , calculated by:

$$I = 100\% \times \frac{1}{N \times H_d} \sum_{i=1}^N |S_i - S_c| \quad (8)$$

where: N is the number of particles; S_i is the displacement vector of particle i ; and S_c is the displacement vector of the sample center of mass. Obviously, the lower the I , the better the lunar soil stratification information is preserved, as shown in Figure 6e. The I increases monotonically as the D increases from 14 to 22 mm, indicating progressive degradation of stratification information.

At smaller D , particle motion space is found to be smaller, relative position transformations are smaller, so I is lower. Furthermore, an approximate linear relationship is evident between I and D . This suggests that the D creates a significant influence on the stratification information.

3.1.2 | Collection Law Analysis

It is important to study the influence of drilling tool parameters on the collection law to avoid clogging. To observe the motion process of the particles into the drill pipe more directly, the initial lunar soil is colored at 10 mm intervals, and the evolution of lunar soil stratification at different D is shown in Figure 7a. It can be divided into three regions: the new sample region, the radial movement region, and the flow region as shown in Figure 7b. First, the particles entering the drill pipe are squeezed and arched to form a new sample region. Subsequently, the lower particles release the pressure by pushing the upper particles in a peripheral direction, distributing them near the inner wall and forming a radial motion region. Then, there is a velocity difference between particles in the new sample region and near the drill pipe, which causes the particles' convection and forms a flow region. The flow region is gradually elongated under the combined effect of the drill tool and the new sample region. This phenomenon becomes more pronounced in the upper segment of the sample, which can be confirmed in Tang et al. in their work [11]. Its specific reasons and manifestations are discussed in Section 3.1.3.

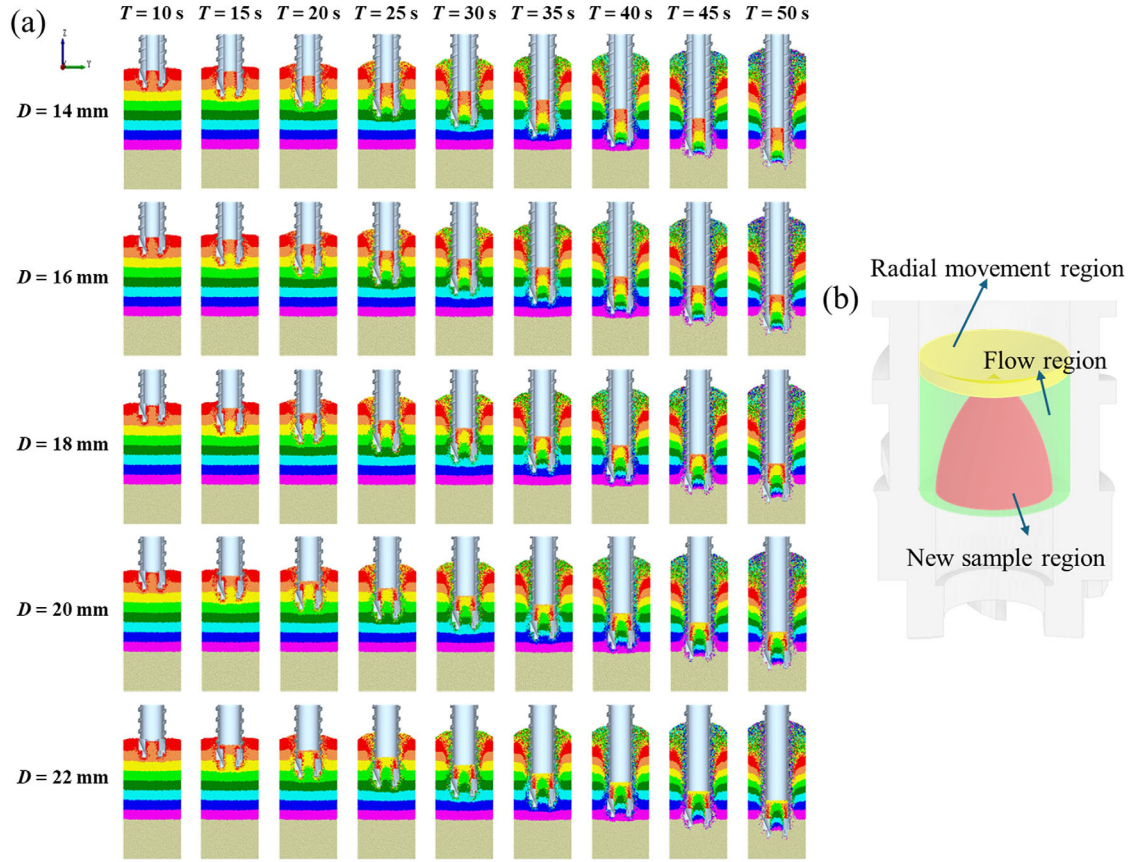


FIGURE 7 | (a) Evolutions of lunar soil stratification at different D ; and (b) three regions of the sample.

Furthermore, the smaller the D , the smaller the motion space, the slower the upper particles move, indicating that the lower particles are pushed less effectively, and their pressure is more difficult to release.

3.1.3 | Mechanism Analysis

Aforementioned results indicate that a larger D will increase m_s and I . To better reveal the underlying mechanism, the speed and force acting on the particles are analyzed. Figure 8a shows the evolutions of particle relative velocity vector v_a based on the motion coordinate system of the drilling tool (i.e., particle Z -direction velocity minus the drilling tool velocity) for obtaining the same sample height. The color represents the relative velocity component in the Z -direction v_{az} . Particles within the new sample region remain at high v_{az} and move upward, the particles' v_{az} in the radial movement region decrease gradually and move toward the inner wall of the drill pipe, and the particles in the flow area rotate in a wave mode, as described in 3.1.2. At larger D , the v_a in the new sample region has a smaller angle to the horizontal and tends to move more towards the inner wall of the drill pipe. This is because the more particles within the drill pipe allow for more collisions with larger motion space, which promotes energy dissipation and more v_{az} reduction. Particle rotation near the drill pipe results from friction and collisions during contact between particle and pipe, which further dissipate energy and reduce v_{az} to nearly zero. The three-dimensional rotational direction of the particles in the flow region is driven by upward pushing of

the particles in the new sample region first. Subsequently, the particles move downwards due to energy dissipation after the collision. The convection induced by this wave-like rotational phenomenon is particularly pronounced at large D , exhibiting the jump-down behavior of the particles as described by Liu et al. [25]. Besides, analysis of the probability distributions of v_{az} of samples in Figure 8b reveals a peak in the curve. With the increase of D , the peak of v_{az} decreases gradually. This is consistent with the rule of change of the average v_{az} of samples in Figure 8c, indicating that when the sample height is the same, the larger the D , the less likely the particles are to have relative motion in the Z direction with the drill pipe (meaning that the particles are not easy to enter the drill pipe).

In addition to v_a , the normal force F_n is another significant factor in sampling. Figure 9a shows the network based on normal force when the same H_s is obtained at different internal diameters. It can be seen that some dendritic networks are formed within the drill pipe. The force chain is weaker farther away from the drilling bit (less than $1\text{e-}4\text{ N}$, shown in white), and becomes stronger at the junction of the drill bit and the drill pipe (between $1\text{e-}4\text{ N}$ and $2\text{E-}4\text{ N}$, shown in light blue) and stronger (greater than $2\text{E-}4\text{ N}$, shown in red) upon closer to the drill bit. Obviously, the smaller the D , the more densely distributed the strong chains which are connected to the inner wall of the drill pipe. Currently, the force chains on both sides readily connect, forming an arch that impedes particle movement. Combined with the probability distribution of the horizontal component of the normal force in Figure 9b and the proportion of the horizontal force component of

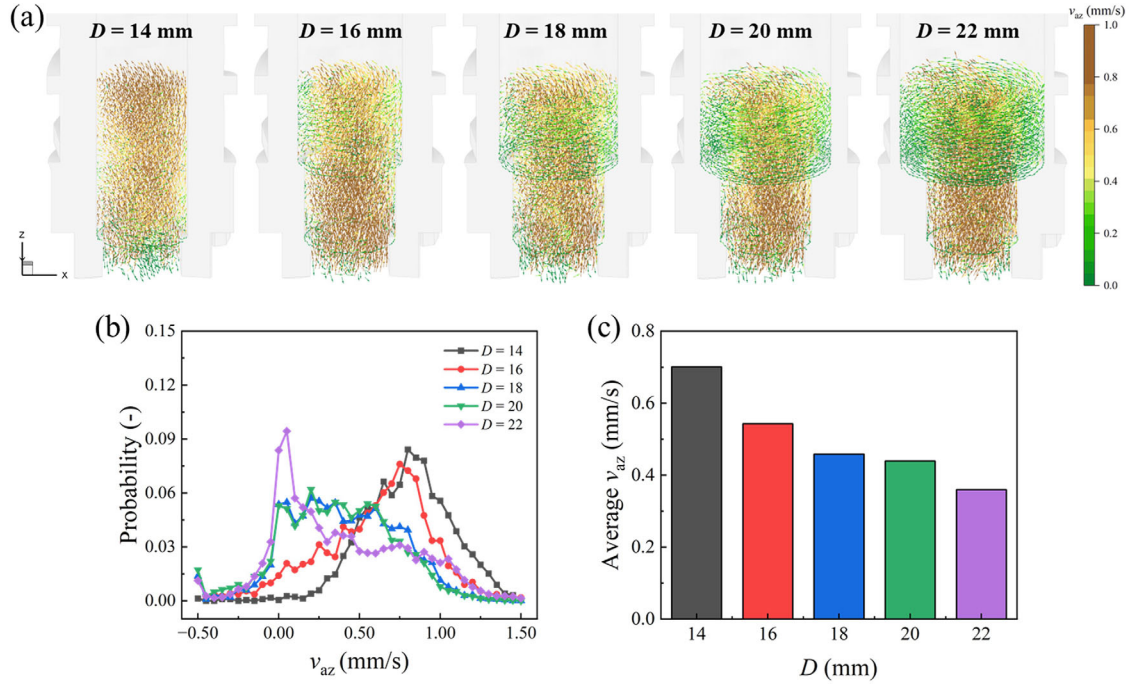


FIGURE 8 | (a) Evolutions of v_a ; probability distributions of v_{az} (b) and average v_{az} (c) of samples at different D .

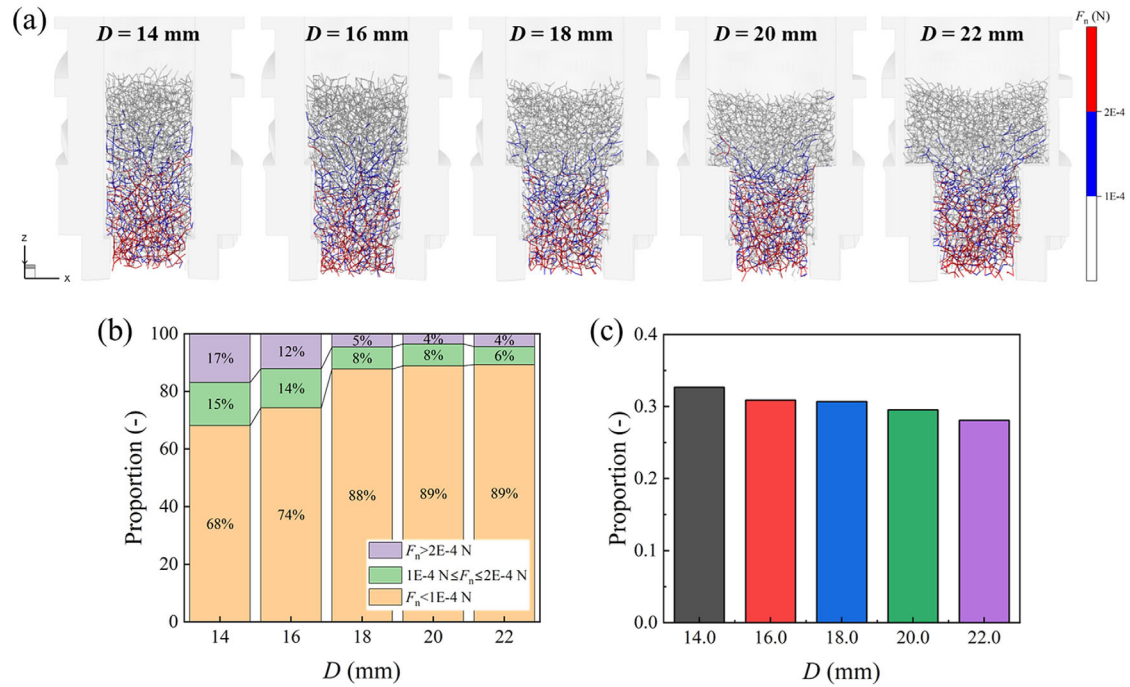


FIGURE 9 | (a) Network based on normal force; proportion of contact force (b) and horizontal force component of the normal force greater than that in the Z-direction (c) at different D .

the normal force larger than that in the Z-direction in Figure 9c, it is clear that the proportion of the strong horizontal component of the normal force gradually increases and plays an important role as the D decreases [11]. This mechanism explains the observed decrease in both sample mass and disturbance rate with decreasing D .

3.2 | Effects of the Friction Coefficient

3.2.1 | Collection Performance Analysis

The effect of friction coefficient μ on m_s and τ_m are analyzed in Figure 10a,b. As can be seen that with the decrease of μ , the m_s

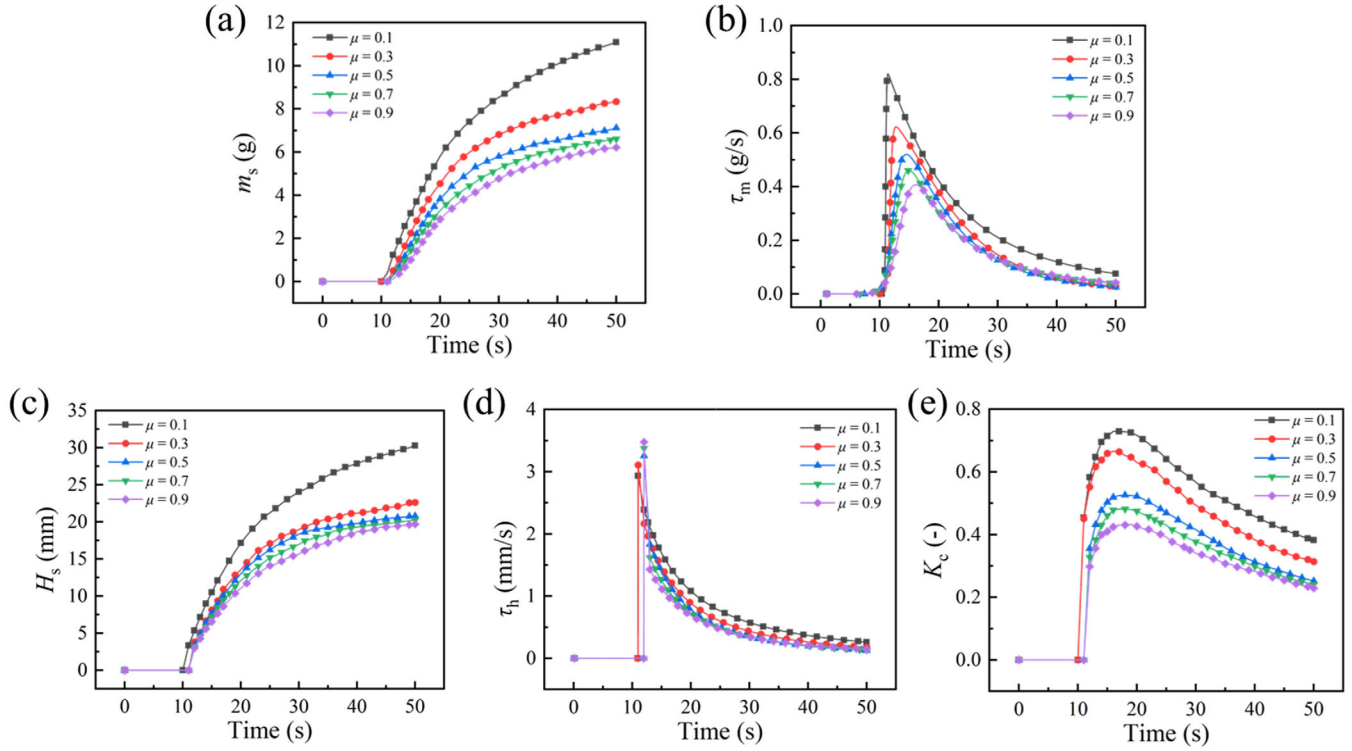


FIGURE 10 | Evolutions of m_s (a), τ_m (b), H_s (c), τ_h (d), and K_c (e) at different μ .

increases because of the change of τ_m . When μ is small, the higher peak of τ_m can be attributed to reduced energy dissipation which allows a greater mass of particles to enter the drill pipe per unit time. At the same time, the reason for the peak of τ_m is closer to the left is that when μ is small, the particles contact the inner wall of the drill pipe early, forming resistance to prevent the movement of particles. Thus, increasing m_s can be achieved by decreasing μ .

Figure 10c–e analyzes the evolutions of H_s , τ_h , and K_c with μ . It can be noticed that as the μ decreases, H_s and τ_h increase. In Stage I, particles form a higher arch at larger μ , as evidenced by elevated peaks in τ_h . When D is 16 mm, the minor discrepancy between the inner diameter of the drill pipe and the drill bit results in a less pronounced particle collapsing and filling behavior, eliminating Stage II. In Stage III, the resistance begins to play a role, and τ_h gradually decreases more at larger μ . Figure 10e shows that the smaller the μ , the higher the K_c . The reason is that when the μ is small, the H_s increases fast, the gap between H_d and H_s gradually becomes smaller, and the K_c becomes higher. The continuous decrease in τ_h over time results in the gap between H_d and H_s becoming progressively larger, so the K_c becoming lower. Consequently, the drill tool with a small μ has been demonstrated to facilitate a greater H_s [22].

The effect of μ on the porosity of the sample is shown in Figure 11a. The porosity curve shows a decreasing trend with decreasing μ , which indicates that particles are more likely to fill the pores through movement in a drill pipe. This filling behavior makes the porosity of the upper and lower layers of the sample more different, and the uniformity is worse. Figure 11b shows the effect of μ on the δ and porosity of the first four layers. Layer 1 displays small δ at the boundaries and large δ in the central region due to the presence of compression zone and rearrangement zone.

This phenomenon is not affected by μ . However, the μ affects the porosity of the subsequent three layers of the sample; the higher the μ , the greater the change of porosity layer by layer, which is consistent with the homogeneity results analyzed above. As shown in Figure 11c, a larger μ reduces particle mobility, resulting in a decrease in the relative displacement between the particles and the sample center of mass, resulting in a lower I . It should be noted that when the μ increases beyond 0.5, its effects on porosity, homogeneity, and I become negligible.

3.2.2 | Collection Law Analysis

The evolution of the particle position during collection at different μ is shown in Figure 12. The effect of the small μ is mainly reflected in that the arch height of the new sample region is lower, the radial movement region is smaller, and the flow region is more elongated. This suggests that the lower particles exert only limited upward force on the upper particles. At the same time, as the μ decreases, the energy loss of the particles decreases, resulting in the smaller velocity difference of the particles in the flow region, so the relative position of the particles is maintained well. Similarly to porosity, there is little difference in the evolution of the particle position when the μ is greater than 0.5. This requires further analysis of the effect of the μ on the samples in terms of particle velocity and force.

3.2.3 | Mechanism Analysis

Figure 13a shows the evolution of particle relative velocity vector v_a . The v_{az} in the new sample region decreases gradually with an

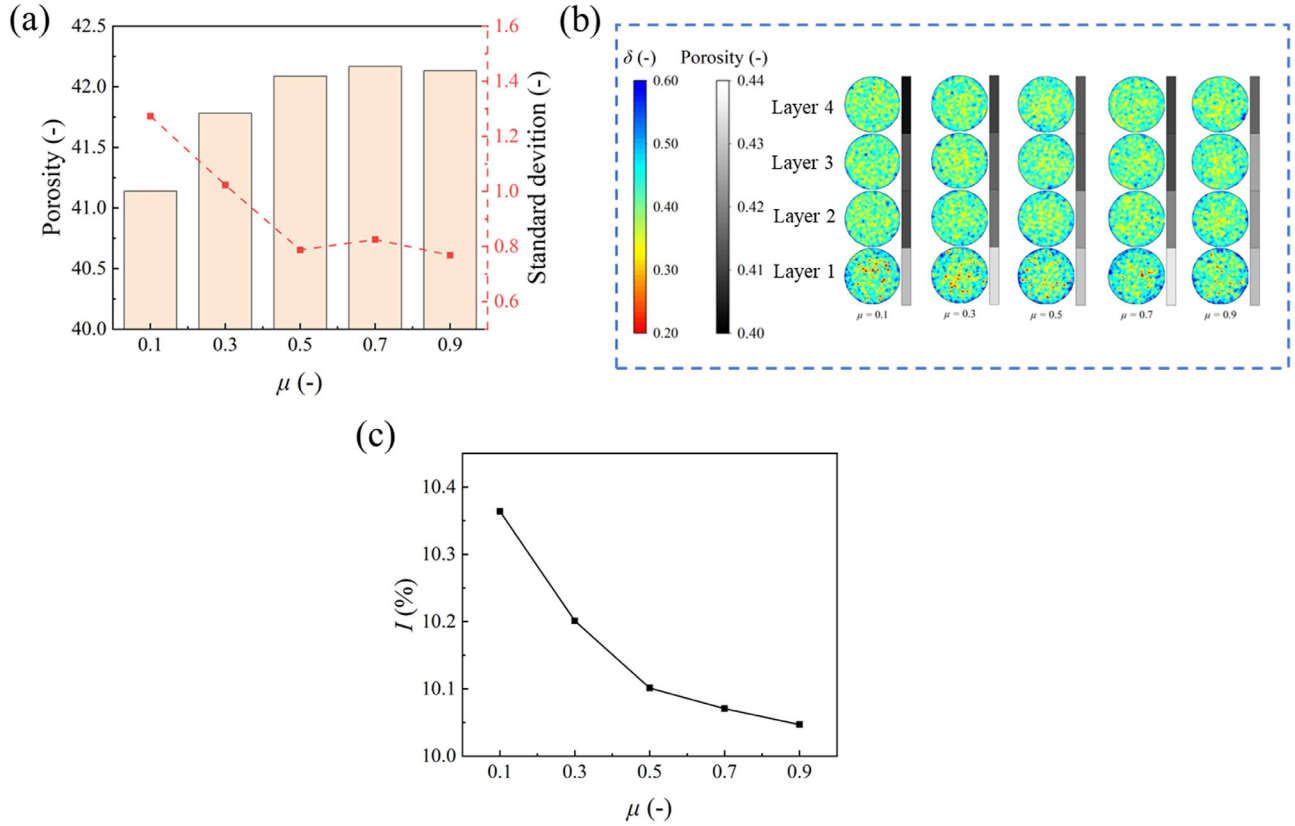


FIGURE 11 | (a) Analysis of porosity and homogeneity at different μ ; (b) δ and porosity of different layers at different μ ; and (c) evolution of I at different μ .

increase in μ . This indicates that a larger μ intensifies the energy dissipation of particles and increases the number of particles with v_{az} close to zero, which is manifested as the increase of rotating particles. On the other hand, pronounced wave-like particle circulation is observed when the μ is appropriate. Because when the μ is too small or too large, the difference of v_{az} in the flow region is small, and the wave-like rotation is not obvious. Figure 13b shows that the v_{az} of the sample corresponding to its peak gradually decreases with the increase of the μ , which is the same as the variation rule of the average v_{az} of the sample in Figure 13c. This means that at the same height of the sample, the larger the μ , the less likely the particles to move to the drill pipe in the Z direction (meaning that the particles are less likely to enter the drill pipe). However, when the μ increases beyond 0.5, it has less effect on the peak of v_{az} as well as on the average v_{az} . This also explains why their m_s and I are always close to each other.

Figure 14a shows the network based on the normal force at different μ . As can be seen, the red force chain is dispersed as the μ decreases. The probability distribution of the horizontal component of the normal force in Figure 14b and the proportion of the horizontal force component of the normal force greater than that in the Z-direction in Figure 14c indicate that the ratio of the strong horizontal component of the normal force chain gradually increases with the μ , and the orientation of the force chain is gradually changed to the horizontal direction [23]. The strong force arch generated at larger μ inhibits particle motion, resulting in lower m_s and I .

3.3 | Effects of the Height of the Complement

3.3.1 | Collection Performance Analysis

Figure 15a,b shows that both m_s and τ_m gradually increase as the height of the complement interior the drill bit H decreases. This is because the complement mainly influences the sampling process by affecting the state of the particles as they enter the drill pipe. Initially, the drill bit continues to do work on the particles while in contact with it. After contacting the inner wall of the drill bit, the diameter difference between the drill bit and the complement makes a modification of particle motion direction, resulting in energy loss and an incremental transition from disordered to ordered states. When the H is larger, the distance of the particles contacting the inner wall of the drill bit is shorter, the particles are still in disordered state after entering the drill pipe, the collision between particles increases, and the energy loss is fast, thereby reducing τ_m . However, when the complement height is less than 8 mm, the particle ordering distance is long enough to make the H have a small effect on the m_s and τ_m . Thus, increasing m_s can be achieved by decreasing H .

The influences of H on the evolution of H_s , τ_h , and K_c are analyzed in Figure 15c-e. It can be noticed that as the H decreases, the H_s increases. And the H of 16 mm results in the highest peak τ_h , but the τ_h is minimum after that. This is because the greater the H , the less energy loss of particles when they pass through the bit, resulting in a higher initial τ_h . However, the continued sampling

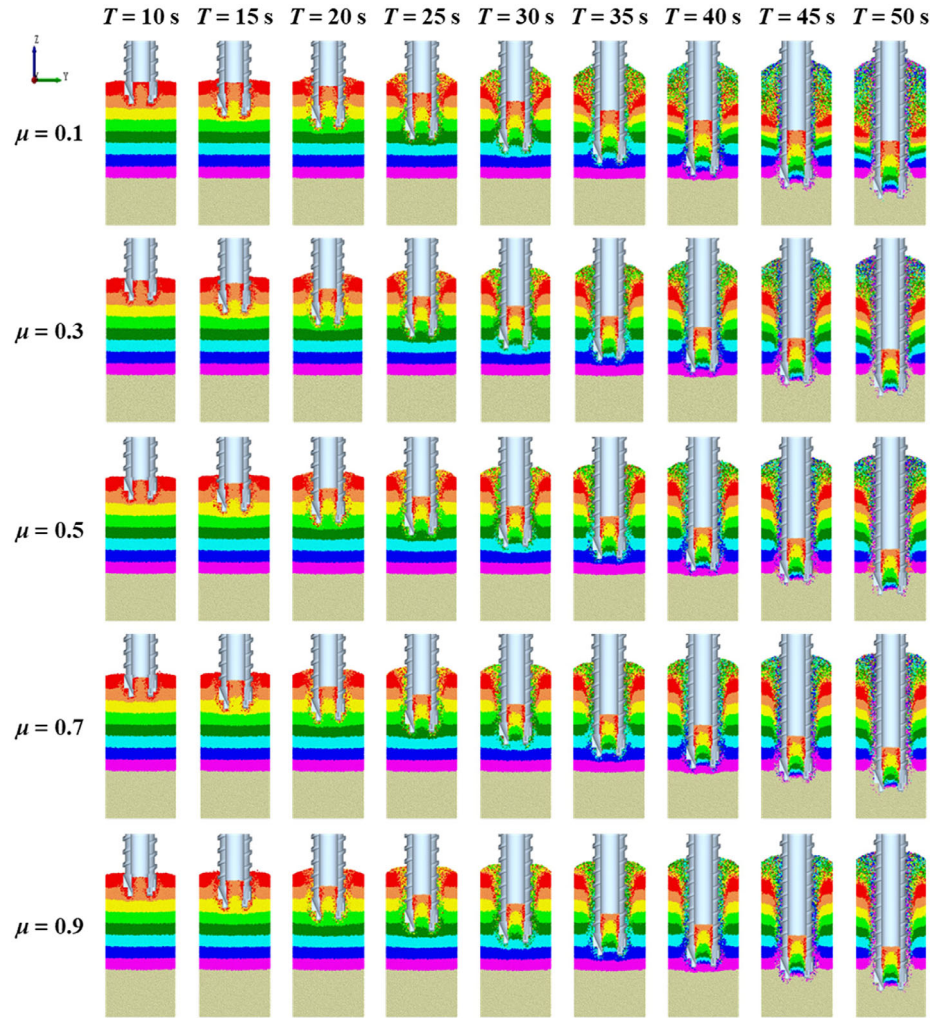


FIGURE 12 | Evolution of lunar soil stratification at different μ .

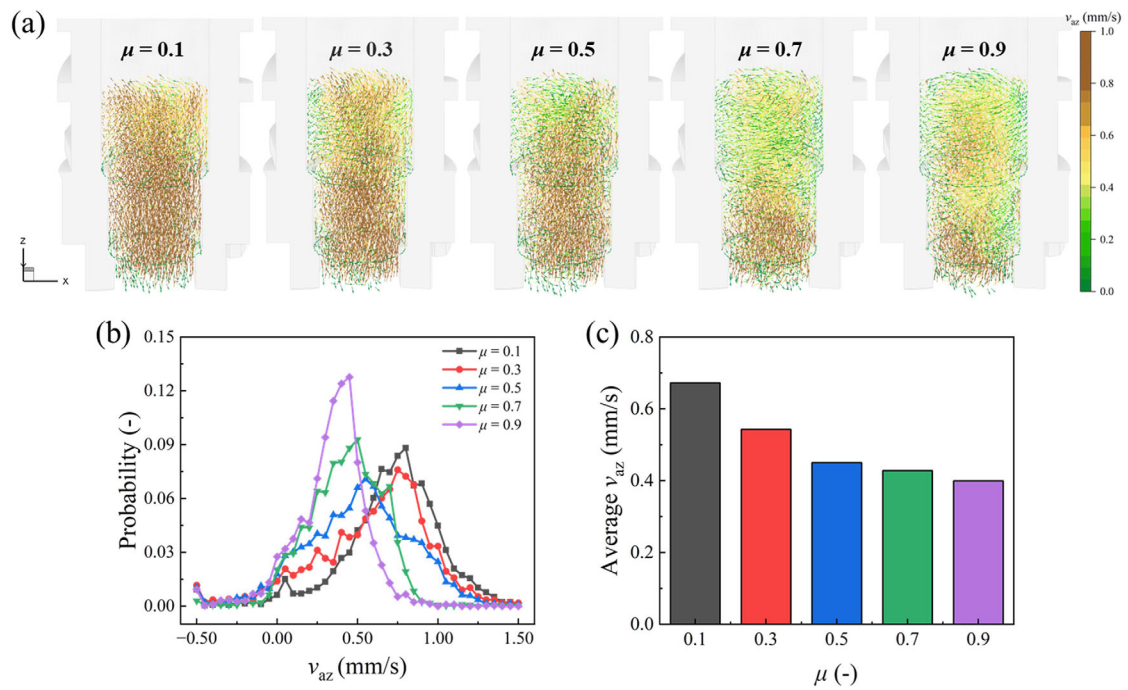


FIGURE 13 | (a) Evolutions of v_{az} ; (b) probability distributions of v_{az} at different μ ; and (c) average v_{az} of the sample at different μ .

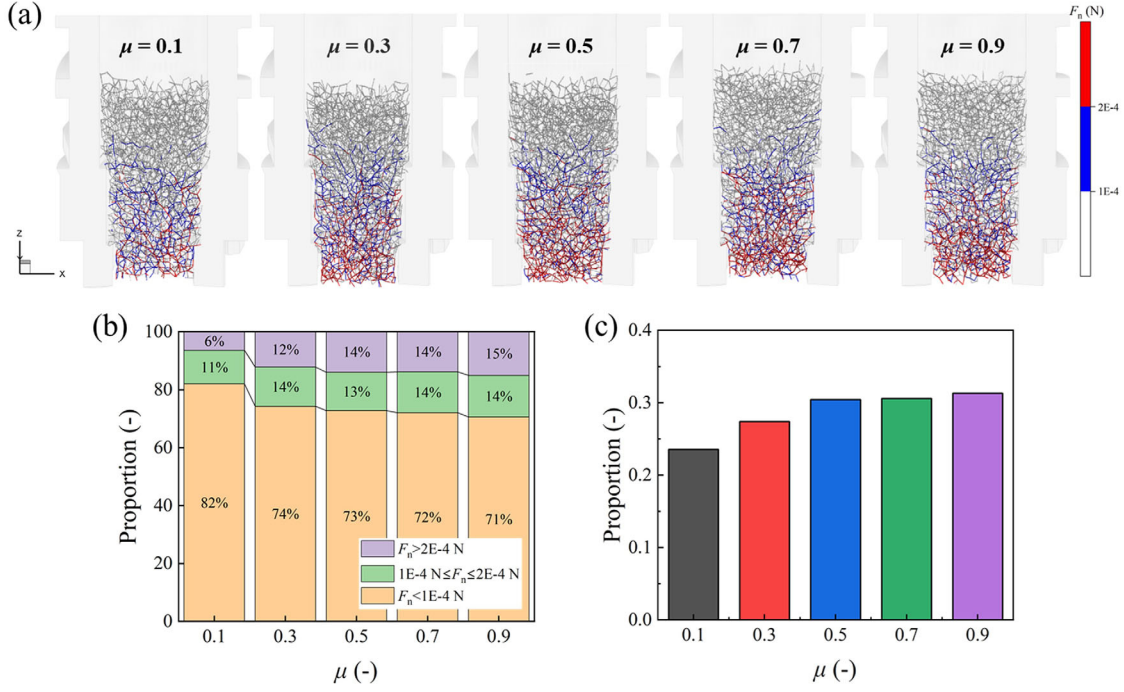


FIGURE 14 | (a) Network of local normal force; proportion of contact force (b) and horizontal force component of the normal force greater than that in the Z-direction (c) at different μ .

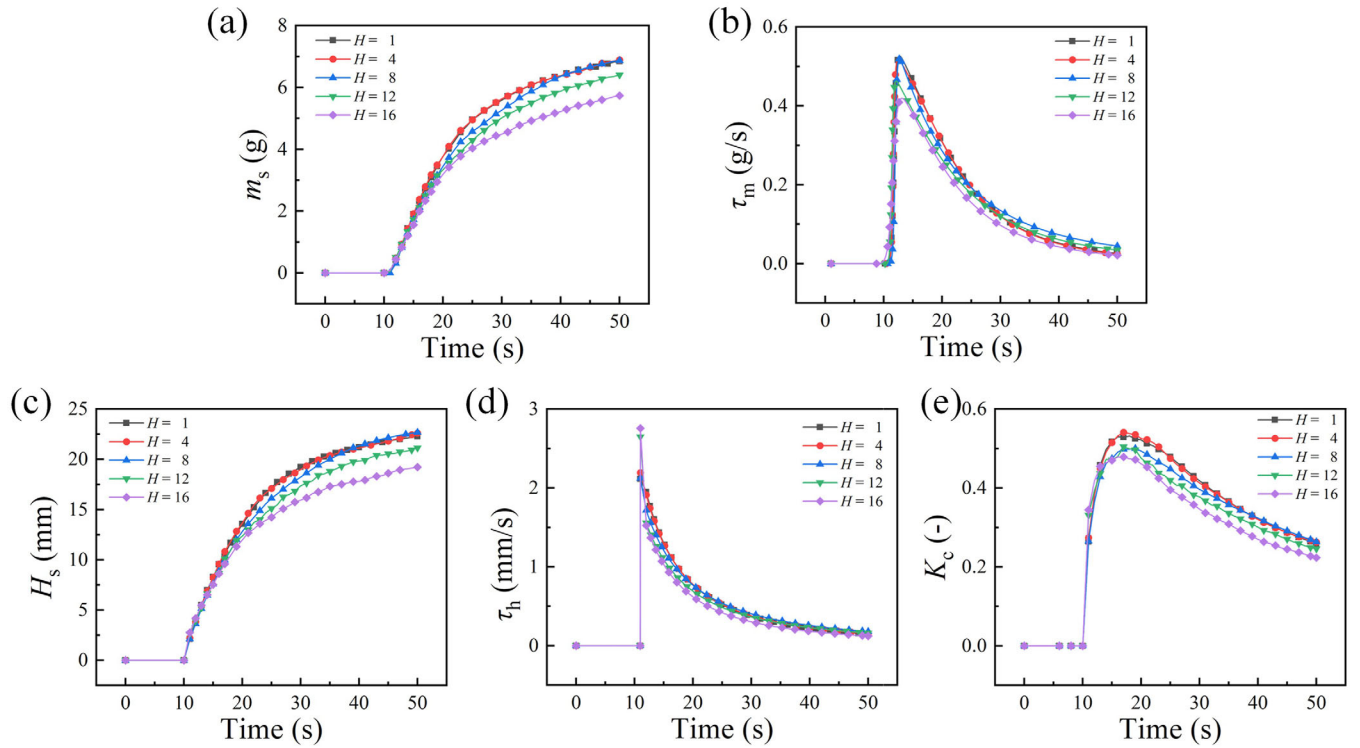


FIGURE 15 | Evolutions of m_s (a), τ_m (b), H_s (c), τ_h (d), and K_c (e) at different H .

caused more particle collisions and increased energy loss, leading to lower τ_h . Caused by the variation of τ_h , the smaller the H , the higher the K_c . As a result, a higher H_s can be achieved by using a drilling pipe of lower H .

Furthermore, an increase in the porosity and δ of the samples is observed with increasing H , as illustrated in Figure 16a. This occurs because the higher the H , the more disordered the particle motion, the more difficult to rearrange the position of particles, so

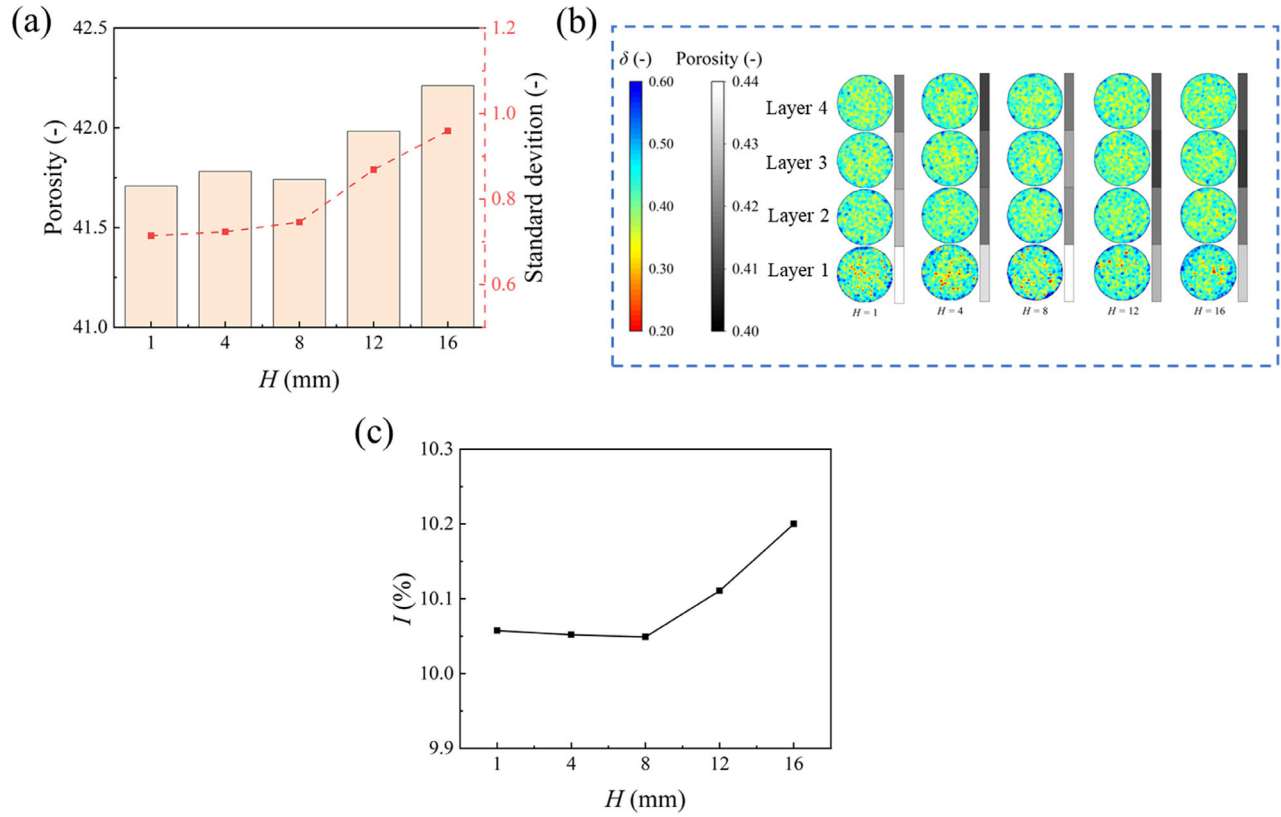


FIGURE 16 | Analysis of porosity and homogeneity (a), δ and porosity of different layers (b), and evolution of I (c) at different H .

the sample porosity is larger and unevenly distributed. Figure 16b shows the effect of H on the δ and porosity of the first four layers. The porosity of Layer 1 increases with the H , demonstrating the presence of turbulence. As observed in Figure 16c, with the increase of the H , the turbulence of particle motion increases the I of the sample and causes deterioration of stratification information. When the H is less than 8 mm, it has less effect on porosity, homogeneity, and I , this is because the particles undergo a long period of regular movement within the drill bit.

3.3.2 | Collection Law Analysis

The effect of H on particle position during collection is shown in Figure 17. The complement exacerbates radial movement of surface particles inside the drill bit. For example, when the H is 1 mm and the time is 10 s, the surface red particles move near the inner wall of the drill bit, and the yellow particles become the new surface particles of the sample. Conversely, when the H is 16 mm, the boundary between the upper and lower layers of particles in the new sample region becomes distorted. This outcome can verify the exacerbation of the turbulence of the particles entering the drill pipe when H is large.

3.3.3 | Mechanism Analysis

The distributions of particle relative velocity v_a in Figure 18a indicate that when H is higher, less energy is lost in the drill bit; the v_{az} of particles in the new sample region is larger, the

collision between particles increases, and the particle movement is disorder, which prevents the subsequent particles from entering. Meanwhile, energy loss accelerated, resulting in a rapid decrease in v_{az} and an increase in the number of rotating particles. However, from the probability distributions of v_{az} of the samples in Figure 18b, one can find that an increase in H brings a decrease in v_{az} , corresponding to the peak of the curve, which is the same as the pattern of change in the average v_{az} of the samples in Figure 18c. When the H is less than 8 mm, its effect on the peak of probability distributions of v_{az} and the average v_{az} is similar, which explains why the evolution of τ_m and τ_h coincide during sampling.

Figure 19a gives the network of normal force for different H . As indicated, at larger H , the disordered particle motion causes the red force chain distribution to converge at the junction of the drill bit and the drill pipe, preventing the sample from increasing. The probability distribution of the horizontal component of the normal force in Figure 19b demonstrates that the ratio of the strong horizontal force gradually increases with the H . From Figure 19c one can find that the larger the H , the easier the strong force arch, resulting in a lower m_s and a higher I .

3.4 | RSM Experimental Design and Statistical Analysis

To predict the sample mass and sample disturbance under different structural parameters, the response surface methodology (RSM)-commonly used for mechanical design and process optimization-was applied [45, 46]. This approach is particularly

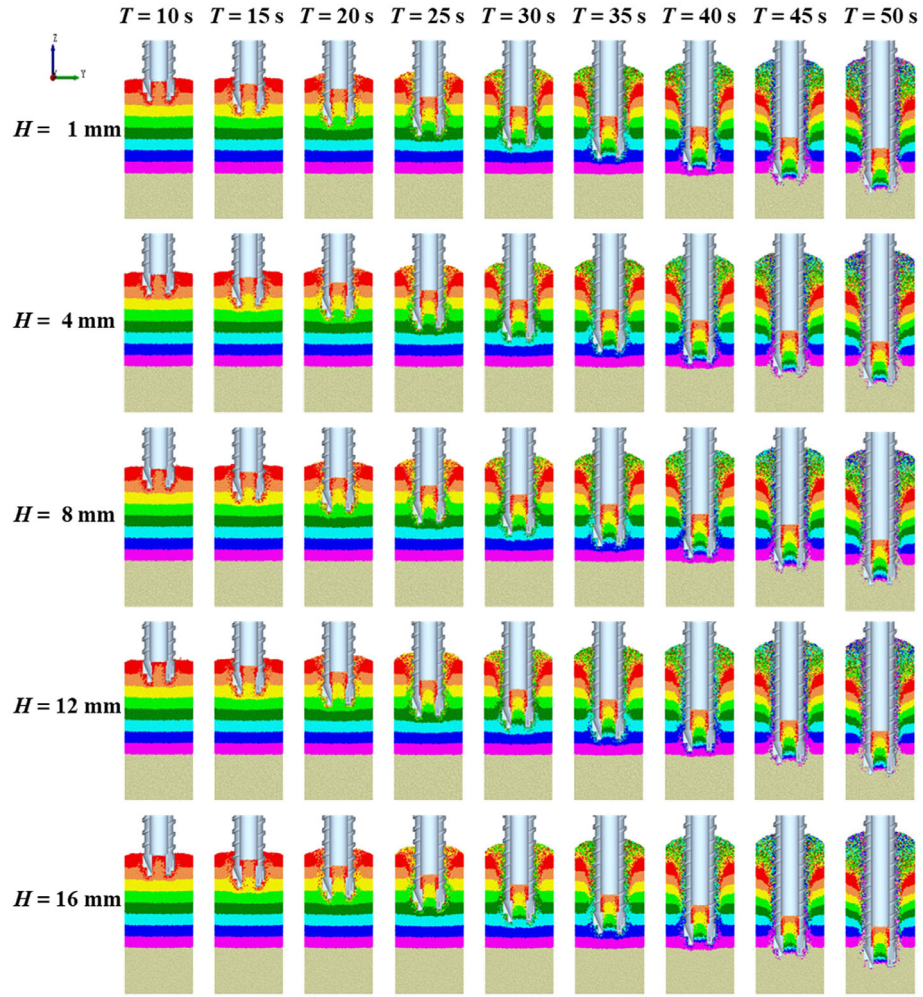


FIGURE 17 | Evolutions of lunar soil stratification at different H .

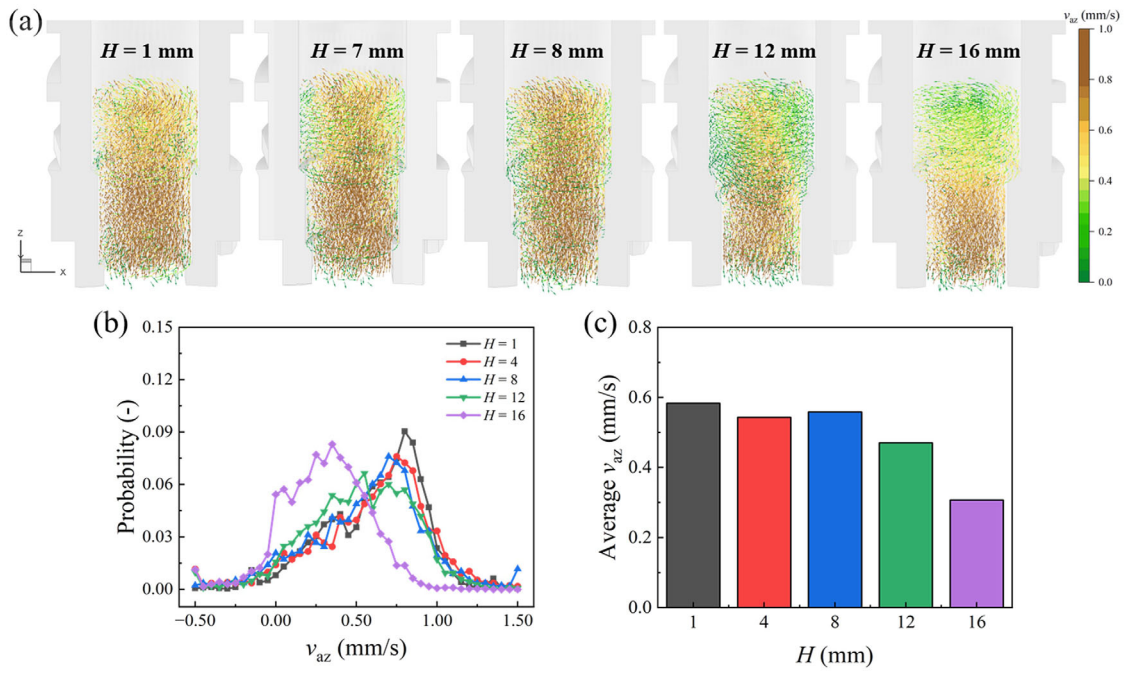


FIGURE 18 | (a) Evolutions of v_{az} ; probability distribution of v_{az} (b) and average v_{az} (c) of sample at different H .

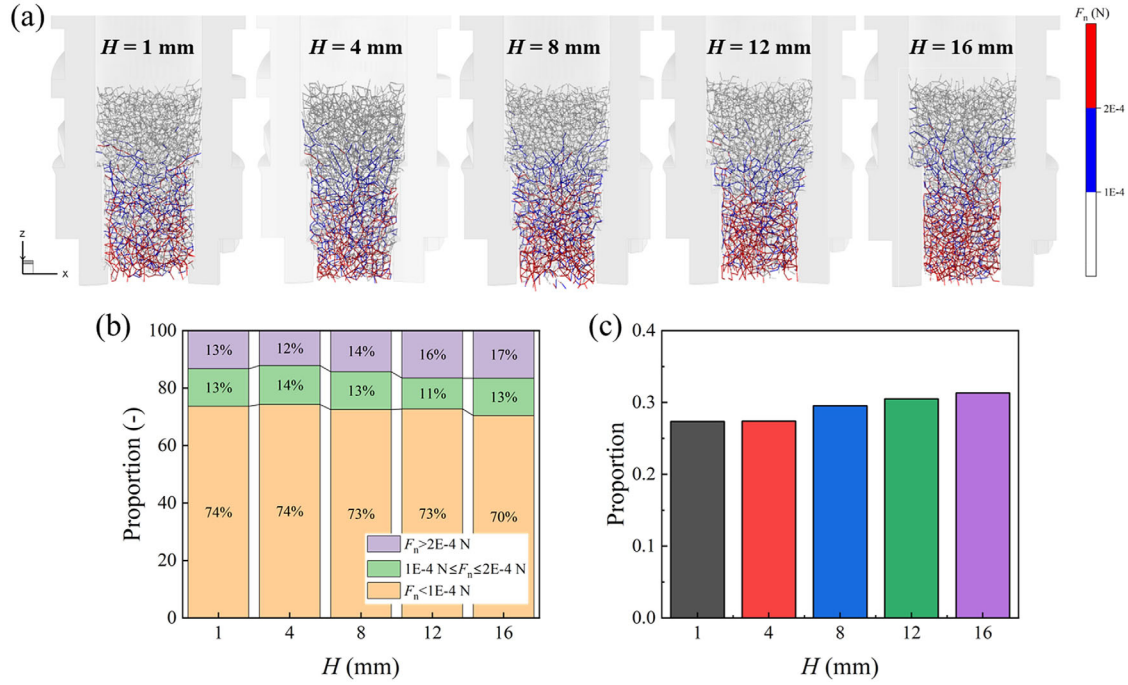


FIGURE 19 | (a) Network of local normal force; proportion of contact force (b) and horizontal force component of the normal force greater than that in the Z-direction (c) at different H .

TABLE 2 | Values of the parametric factors.

Parameters	The inner diameter of the drill pipe, D (mm)	Height of the complement interior the drill bit, H (mm)	The friction coefficient, μ (-)
Lower (-1)	14	1	0.1
Mean (0)	18	8.5	0.5
Upper (1)	22	16	0.9

important for space missions, where testing opportunities are extremely limited. In this part, D , H , and μ are designated as variables, the focused sample mass m_s and the disturbance rate I are designated as response values. The values of the parametric factors are shown in Table 2, while the corresponding simulation results are provided in Table A2 of Appendix A2. Sample mass m_s and disturbance rate I prediction models are established based on quadratic regression fitting, as given by:

$$\begin{cases}
 m_s = * 20l - 16.30556 + 3.54298D - 34.47255H - 0.25665\mu \\
 - 0.80484D \times H - 5.82658^{-3}D \times \mu + 0.036134H \times \mu \\
 - 0.065661D^2 + 31.36457H^2 + 0.012640\mu^2 \\
 I = -15.32127 + 2.02205D + 12.68509H - 0.28803\mu \\
 - 0.76697D \times H + 8.57875^{-3}D \times \mu - 0.033683H \times \mu \\
 - 0.024403D^2 - 0.78625H^2 + 9.80996^{-3}\mu^2
 \end{cases} \quad (9)$$

To determine the significance of each factor and its interaction with the response variable, statistical technique of variance (ANOVA) is used. The ANOVA results are shown in Table 3, indicating that μ has the greatest effect on sample mass m_s , while D is the most significant factor affecting the disturbance rate I .

The p-values are less than 0.0001, indicating the significance of the predictive model. The response models for R^2 are greater than 0.95, implying that the model can predict the sample mass and disturbance rate more accurately within the selected parameter range. Residual normal probability plots and correlation analyses are performed, as detailed in Figure A1 of Appendix A2. It indicates that each prediction model is well adapted and can be used for prediction. The validation of the prediction model is performed by selecting three groups of parameters (D - H - μ), as shown in Figure 20, the errors of sample mass m_s and disturbance rate I predicted by the prediction model with the simulation results verified the accuracy of the prediction models.

4 | Conclusions

The flow and collection of lunar soil inside a hollow drilling tool are numerically modelled by DEM. The effects of drilling tool parameters (the inner diameter of the drill pipe D , the friction coefficient μ , and the height of the complement interior the drill bit H) on the macroscopic properties (sample mass, coring ratio and uniformity, etc.), microscopic properties (pore size and disturbance), and collection law on the samples are investigated.

TABLE 3 | Results of ANOVA for all quadratic models.

Model	<i>F</i> value (<i>D</i>)	<i>F</i> value (<i>H</i>)	<i>F</i> value (μ)	<i>p</i> -value (Model)	<i>p</i> -value (Lack of fit)	<i>R</i> ²
<i>m_s</i>	77.95	8.57	440.47	<0.0001	0.0514	0.9896
<i>I</i>	162.66	0.22	11.27	0.0003	0.7309	0.9643

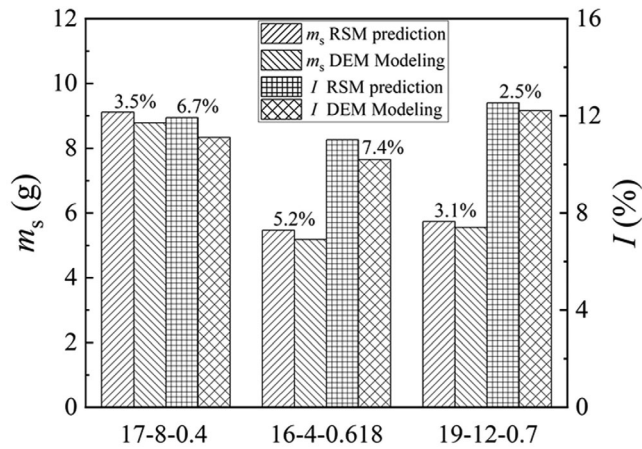


FIGURE 20 | Comparison of sample mass and disturbance rate under three groups of different parameters.

On this basis, the prediction models are established by RSM and DEM with the objectives of sample mass m_s and disturbance rate I and verified. Following conclusions can be drawn.

1. Larger D , smaller μ , and smaller H significantly increase m_s . Smaller D , larger μ , and smaller H can significantly reduce I .
2. The sample can be divided into three regions: the new sample region, the radial movement region, and the flow region. A larger D makes the flow region larger, and the motion of the particles presents a wave-like rotation. The smaller μ makes the energy loss of particles in the new sample area less and the radial movement region smaller. Smaller H makes the particle movement in the new sample region more orderly.
3. For the same sampling height, smaller H makes the particles have a greater average velocity in the Z direction relative to the drilling tool, and the horizontal force arch formed by the normal force between particles is weaker, which together cause m_s to increase and I to decrease. However, smaller D and larger μ obtain smaller m_s and I because the force arch impedes the particle motion too strongly.
4. The quadratic regression models for sample mass and disturbance rate exhibited high predictive accuracy ($R^2 > 0.95$). These prediction models provide a reliable solution for forecasting sampling outcomes under varying drilling tool parameters, balancing sample mass and stratification preservation.

The analysis of lunar soil is of paramount significance for unraveling the Moon's formation and evolutionary history. Although the present DEM model provides valuable insights into lunar soil collection within a hollow drilling tool, several limitations

should be acknowledged. The lunar regolith is represented using simplified particle shapes and material properties, and the simulations consider only a limited range of drilling tool parameters under idealized geological conditions. Factors such as regolith stratification, rock fragment content, density variations, and more complex operational conditions are not explicitly incorporated. To transit simulations from fundamental analysis to mission-ready predictive tools, future work should systematically integrate these geological and operational factors, including real-time parameter adjustment and more realistic lunar environment.

Author Contributions

Ge Yu: conceptualization, data curation, formal analysis, investigation, methodology, writing – original draft. **Meng Li:** conceptualization, data curation, resources, validation, writing – review and editing. **Hongyi Ge:** formal analysis, methodology, visualization, writing – review and editing. **Jiangyuan Zhang:** data curation, validation, writing – review and editing. **Ju Wang:** Formal analysis, Validation, Writing – review and editing. **Kefan Cao:** Methodology, Writing – review and editing. **Xizhong An:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – review and editing.

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Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data Availability Statement

Data will be made available on request.

References

1. Y. Gumulya, L. Zea, and A. Kaksonen, "In Situ Resource Utilisation: The Potential for Space Biomining," *Minerals Engineering* 176 (2022): 107288, <https://doi.org/10.1016/j.mineng.2021.107288>.
2. S. Hu, H. He, J. Ji, et al., "A Dry Lunar Mantle Reservoir for Young Mare Basalts of Chang'e-5," *Nature* 600, no. 7887 (2021): 49–53, <https://doi.org/10.1038/s41586-021-04107-9>.
3. Q. Yuan, M. Li, S. J. Desch, et al., "Moon-Forming Impactor as a Source of Earth's Basal Mantle Anomalies," *Nature* 623, no. 7985 (2023): 95–99, <https://doi.org/10.1038/s41586-023-06589-1>.
4. C. Bao, D. Zhang, Q. Wang, Y. Cui, and P. Feng, "Lunar in Situ Large-Scale Construction: Quantitative Evaluation of Regolith Solidification Techniques," *Engineering* 39 (2024): 204–221, <https://doi.org/10.1016/j.eng.2024.03.004>.

5. Q. Cui, T. Wang, G. Gu, et al., "Ultraviolet and Thermal Dual-Curing Assisted Extrusion-Based Additive Manufacturing of Lunar Regolith Simulant for In-Site Construction on the Moon," *Construction and Building Materials* 425 (2024): 136010, <https://doi.org/10.1016/j.conbuildmat.2024.136010>.
6. Y. Yao, L. Wang, X. Zhu, et al., "Extraterrestrial Photosynthesis by Chang'E-5 Lunar Soil," *Joule* 6, no. 5 (2022): 1008–1014, <https://doi.org/10.1016/j.joule.2022.04.011>.
7. C. K. Shearer, F. M. McCubbin, S. Eckley, et al., "Apollo Next Generation Sample Analysis (ANGSA): An Apollo Participating Scientist Program to Prepare the Lunar Sample Community for Artemis," *Space Science Reviews* 220, no. 6 (2024): 62, <https://doi.org/10.1007/s11214-024-01094-x>.
8. S. Okabayashi, T. Yokoyama, T. Hirata, K. Terakado, and E. M. Galimov, "Iron Isotopic Composition of Very Low-Titanium Basalt Deduced From the Iron Isotopic Signature in Luna 16, 20, and 24 Soils," *Geochimica et Cosmochimica Acta* 269 (2020): 1–14, <https://doi.org/10.1016/j.gca.2019.10.020>.
9. J. Zhu, M. Zou, Y. Liu, K. Gao, B. Su, and Y. Qi, "Measurement and Calibration of DEM Parameters of Lunar Soil Simulant," *Acta Astronautica* 191 (2022): 169–177, <https://doi.org/10.1016/j.actaastro.2021.11.009>.
10. B. M. Willman, W. W. Boles, D. S. McKay, and C. C. Allen, "Properties of Lunar Soil Simulant JSC-1," *Journal of Aerospace Engineering* 8, no. 2 (1995): 77–87, [https://doi.org/10.1061/\(ASCE\)0893-1321\(1995\)8:2\(77\)](https://doi.org/10.1061/(ASCE)0893-1321(1995)8:2(77)).
11. D. Tang, X. Gao, D. Zhao, S. Jiang, and Z. Deng, "Analyses of Sampling Disturbance of the Lunar Surface in Direct Push Sampling Method," *IEEE Access* 6 (2018): 48656–48663, <https://doi.org/10.1109/ACCESS.2018.2867950>.
12. Y. PANG, T. ZHANG, H. WEI, et al., "Influence of Lunar Regolith Compressibility on Sampling Performance of Thick Wall Spiral Drills," *Chinese Journal of Aeronautics* 36, no. 2 (2023): 350–362, <https://doi.org/10.1016/j.cja.2021.11.017>.
13. Y. Zhang, R. Wu, Z. Long, Z. Zhao, and L. Wang, "Experimental Investigation and Theoretical Modeling of Cutting Mechanics Using Serrated Tines in Drilling Lunar Regolith Simulant," *Advances In Space Research* 74 (2024): 3925–3944, <https://doi.org/10.1016/j.asr.2024.06.063>.
14. J. Xu, J. Liu, S. Yu, K. Xu, and T. Zhang, "Real-Time Temperature Prediction of Lunar Regolith Drilling Based on ATT-Bi-LSTM Network," *International Journal of Heat and Mass Transfer* 18 (2024): 124783, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124783>.
15. J. Xu, X. Yuan, Y. Zhang, et al., "Real-Time Prediction of Drilling Forces Inside Lunar Regolith Based on Recurrent Neural Networks," *Acta Astronautica* 201 (2022): 259–273, <https://doi.org/10.1016/j.actaastro.2022.08.051>.
16. Y. Tian, D. Tang, Z. Deng, S. Jiang, and Q. Quan, "Drilling Power Consumption and Soil Conveying Volume Performances of Lunar Sampling Auger," *Chinese Journal of Mechanical Engineering* 28, no. 3 (2015): 451–459, <https://doi.org/10.3901/CJME.2015.0301.021>.
17. T. Zhang and X. Ding, "Drilling Forces Model for Lunar Regolith Exploration and Experimental Validation," *Acta Astronautica* 131 (2017): 190–203, <https://doi.org/10.1016/j.actaastro.2016.11.035>.
18. Q. Quan, J. Tang, F. Yuan, S. Jiang, and Z. Deng, "Drilling Load Modeling and Validation Based on the Filling Rate of Auger Flute in Planetary Sampling," *Chinese Journal of Aeronautics* 30 (2017): 434–446, <https://doi.org/10.1016/j.cja.2016.05.003>.
19. M. Jiang, Z. Shen, and C. Thornton, "Microscopic Contact Model of Lunar Regolith for High Efficiency Discrete Element Analyses," *Computers and Geotechnics* 54 (2013): 104–116, <https://doi.org/10.1016/j.compgeo.2013.07.006>.
20. J. Li, L. Wang, C. Feng, M. Wen, and Y. Zhang, "Penetration and Resistance Characteristics of Lunar Regolith Simulant Drilling Using a Coupled MPM-CDEM Approach," *Journal of Rock Mechanics and Geotechnical Engineering* 25 (2025): 7755, <https://doi.org/10.1016/j.jrmge.2025.01.029>.
21. A. Luo, Y. Cui, J. Nie, and G. Wang, "Effects of Adhesion and Particle Shape on Mechanical Behaviors of Lunar Regolith Under Low Stress Condition-3D DEM Study," *Computers and Geotechnics* 175 (2024): 106661, <https://doi.org/10.1016/j.compgeo.2024.106661>.
22. H. Wang, Q. Ma, and Q. Wu, "Damage Constitutive Model of Lunar Soil Simulant Geopolymer Under Impact Loading," *Journal of Rock Mechanics and Geotechnical Engineering* 16, no. 3 (2024): 1059–1071, <https://doi.org/10.1016/j.jrmge.2023.04.025>.
23. X. Gao, D. Tang, H. Yue, S. Jiang, and Z. Deng, "Influence of Friction on Sampling Disturbance of Lunar Surface in Direct Push Sampling Method Based on DEM," *Advances in Space Research* 59 (2017): 3036–3044, <https://doi.org/10.1016/j.asr.2017.03.029>.
24. Z. Wang, J. Li, T. Yu, Q. Cheng, F. Li, and Z. Li, "Sampling Dynamic Analysis and Discrete Element Simulation on a Deep Layer Regolith Drill for Extraterrestrial Celestial Bodies," *Acta Astronautica* 225 (2024): 141–157, <https://doi.org/10.1016/j.actaastro.2024.09.012>.
25. T. Liu, J. Zhou, L. Liang, Z. Bai, and Y. Zhao, "Effect of Drill Bit Structure on Sample Collecting of Lunar Soil Drilling," *Advances in Space Research* 68 (2021): 134–152, <https://doi.org/10.1016/j.asr.2021.02.045>.
26. T. Liu, J. Zhou, L. Liang, Y. Zhao, and D. Cao, "Investigation of Effect of Stones on Lunar Regolith Drilling Using DEM," *Computers and Geotechnics* 131 (2021): 103958, <https://doi.org/10.1016/j.compgeo.2020.103958>.
27. H. Chen, L. Li, Y. Cui, and S. Jiang, "A Novel Lunar Soil Coring Approach Based on Particle Unidirectional Flow Effect to Reduce Drag and Increase Efficiency," *Advances in Space Research* 68 (2021): 117–133, <https://doi.org/10.1016/j.asr.2021.02.044>.
28. L. Zhang, L. Wang, Q. Sun, J. Badal, and Q. Chen, "Multi-Objective Design Optimization of Clam-Inspired Drilling into the Lunar Regolith," *Acta Geotechnica* 19 (2024): 1379–1396, <https://doi.org/10.1007/s11440-023-02119-5>.
29. H. Li, Y. Gao, Y. Zhou, and C. Zhou, "Performance Analysis of a Bucket Drum for Lunar Regolith Collection in Lunar Base Construction," *Automation in Construction* 166 (2024): 105617, <https://doi.org/10.1016/j.autcon.2024.105617>.
30. J. Tang, Q. Quan, S. Jiang, et al., "Experimental Investigation on Flowing Characteristics of Flexible Tube Coring in Lunar Sampling Missions," *Powder Technology* 326 (2018): 16–24, <https://doi.org/10.1016/j.powtec.2017.12.013>.
31. M. Li, Z. Li, C. Li, et al., "Numerical Insights on the Combustion Characteristics in the Low-Carbon Blast Furnace Raceway with Hydrogen-Rich Gas Injections," *International Journal of Hydrogen Energy* 100 (2025): 341–354, <https://doi.org/10.1016/j.ijhydene.2024.12.305>.
32. Q. Zou, D. Wu, X. Yu, M. Li, X. An, and J. Jie, "Elaborating Cu-Network Structure of W@20Cu Composites with Cu-Coated W Powders via Oscillating Pressure Sintering," *Journal of Alloys and Compounds* (2025): 178629, <https://doi.org/10.1016/j.jallcom.2025.178629>.
33. F. Fleissner, T. Gaugele, and P. Eberhard, "Applications of the Discrete Element Method in Mechanical Engineering," *Multibody System Dynamics* 18 (2007): 81–94, <https://doi.org/10.1007/s11044-007-9066-2>.
34. S. Jiang, Y. Ye, Y. Tan, et al., "Discrete Element Simulation of Particle Motion in Ball Mills Based on Similarity," *Powder Technology* 335 (2018): 91–102, <https://doi.org/10.1016/j.powtec.2018.05.012>.
35. M. Sakai, M. Abe, Y. Shigeto, et al., "Verification and Validation of a Coarse Grain Model of the DEM in a Bubbling Fluidized Bed," *Chemical Engineering Journal* 244 (2014): 33–43, <https://doi.org/10.1016/j.cej.2014.01.029>.
36. J. Nie, Y. Cui, K. Senetakis, et al., "Predicting Residual Friction Angle of Lunar Regolith Based on Chang'e-5 Lunar Samples," *Science*

Bulletin 68, no. 7 (2023): 730–739, <https://doi.org/10.1016/j.scib.2023.03.019>.

37. C. Li, H. Hu, M.-F. Yang, et al., “Nature of the Lunar Far-Side Samples Returned by the Chang’E-6 Mission,” *National Science Review* 11, no. 11 (2024): 328, <https://doi.org/10.1093/nsr/nwae328>.

38. R. F. Scott, “Failure,” *Géotechnique* 37, no. 4 (1987): 423–466, <https://doi.org/10.1016/j.cej.2014.01.029>.

39. L. T. Zhu, Y. N. Yang, D. T. Pan, and Z. H. Luo, “Capability Assessment of Coarse-Grid Simulation of Gas-Particle Riser Flow Using Sub-Grid Drag Closures,” *Chemical Engineering Science* 213 (2020): 115410, <https://doi.org/10.1016/j.ces.2019.115410>.

40. H. Lei, L. T. Zhu, and Z. H. Luo, “CFD-DEM Study of Reactive Gas-Solid Flows with Cohesive Particles in a High Temperature Polymerization Fluidized Bed,” *Chemical Engineering Science* 268 (2023): 118437, <https://doi.org/10.1016/j.ces.2022.118437>.

41. Q. Wu, C. Qiao, W. Fan, et al., “Insights into Surface Chemistry Induced Powder Layer Characteristic Evolutions in Additive Manufacturing,” *Powder Technology* 407 (2022): 117635, <https://doi.org/10.1016/j.powtec.2022.117635>.

42. J. Tang, X. Chen, H. Zhang, et al., “Soil Flowing Characteristics during Flexible Tube Coring for Lunar Exploration: Modeling and Verification,”

Acta Astronautica 212 (2023): 387–397, <https://doi.org/10.1016/j.actaastro.2023.07.041>.

43. M. Li, L. Zhang, C. Li, et al., “Discrete Element Method–Volume of Fluid Simulation of Drawdown and Dispersion for Floating Particles in Stirred Tanks: Influences of Impeller Parameters,” *Industrial & Engineering Chemistry Research* 63 (2024): 10353–10372, <https://doi.org/10.1021/acs.iecr.4c00935>.

44. Y. Wu, M. Li, J. U. Wang, et al., “Powder-Bed-Fusion Additive Manufacturing of Molybdenum: Process Simulation, Optimization, and Property Prediction,” *Additive Manufacturing* 58 (2022): 103069, <https://doi.org/10.1016/j.addma.2022.103069>.

45. C. Jiang, D. Gou, C. Li, et al., “Crushing Characteristics and Performance Evaluation of Iron Ore in a Cone Crusher: A Numerical Study,” *Minerals Engineering* 204 (2023): 108429, <https://doi.org/10.1016/j.mineng.2023.108429>.

46. Y. Lu, M. Huang, C. Zhang, B. Wang, L. Peng, and W. Wei, “Optimization of Defoaming-Flocculation-Dewatering Indices of Earth Pressure Balance (EPB) Shield Muck Using Response Surface Methodology and Desirability Approach,” *Journal of Rock Mechanics and Geotechnical Engineering* 17, no. 2 (2025): 1134–1148, <https://doi.org/10.1016/j.jrmge.2024.04.001>.

Appendix A

A.1 | AOR Test

TABLE A1 | Parameter combinations and numerical testing results for AOR.

No.	Particle-particle			Particle-drilling tool			AOR (°)
	Restitution coefficient, -	Static friction coefficient, -	Rolling friction coefficient, -	Restitution coefficient, -	Static friction coefficient, -	Rolling friction coefficient, -	
1	0.7	0.5	0.155	0.3	0.9	0.155	46.78
2	0.7	0.5	0.3	0.5	0.5	0.01	50.3
3	0.3	0.5	0.155	0.7	0.9	0.155	49.67
4	0.5	0.5	0.155	0.5	0.5	0.155	50.23
5	0.5	0.1	0.155	0.5	0.1	0.3	32.25
6	0.3	0.5	0.01	0.5	0.5	0.01	35.61
7	0.5	0.1	0.155	0.5	0.9	0.3	33.52
8	0.7	0.5	0.01	0.5	0.5	0.3	37.93
9	0.5	0.9	0.01	0.5	0.1	0.155	34.76
10	0.5	0.1	0.155	0.5	0.1	0.01	34.2
11	0.5	0.5	0.3	0.3	0.5	0.01	52.96
12	0.5	0.9	0.3	0.5	0.9	0.155	51.6
13	0.7	0.9	0.155	0.3	0.5	0.155	47.01
14	0.7	0.9	0.155	0.7	0.5	0.155	47.13
15	0.7	0.5	0.155	0.7	0.9	0.155	46.9
16	0.5	0.5	0.3	0.7	0.5	0.01	52.14
17	0.5	0.5	0.01	0.3	0.5	0.01	35.12
18	0.3	0.9	0.155	0.3	0.5	0.155	49.42
19	0.5	0.5	0.3	0.3	0.5	0.3	52.1
20	0.7	0.5	0.155	0.7	0.1	0.155	47.6

(Continues)

TABLE A1 | (Continued)

No.	Particle-particle			Particle-drilling tool			AOR (°)
	Restitution coefficient, -	Static friction coefficient, -	Rolling friction coefficient, -	Restitution coefficient, -	Static friction coefficient, -	Rolling friction coefficient, -	
21	0.5	0.1	0.3	0.5	0.9	0.155	30.49
22	0.3	0.5	0.3	0.5	0.5	0.01	52.01
23	0.5	0.1	0.01	0.5	0.1	0.155	27
24	0.5	0.9	0.3	0.5	0.1	0.155	51.6
25	0.7	0.1	0.155	0.7	0.5	0.155	30.29
26	0.7	0.1	0.155	0.3	0.5	0.155	34.49
27	0.5	0.9	0.155	0.5	0.1	0.01	46.87
28	0.5	0.9	0.01	0.5	0.9	0.155	35.61
29	0.7	0.5	0.155	0.3	0.1	0.155	46.23
30	0.3	0.5	0.155	0.7	0.1	0.155	47.83
31	0.5	0.5	0.155	0.5	0.5	0.155	49.54
32	0.3	0.5	0.155	0.3	0.1	0.155	49.04
33	0.5	0.5	0.155	0.5	0.5	0.155	50.3
34	0.5	0.9	0.155	0.5	0.9	0.01	47.13
35	0.3	0.5	0.3	0.5	0.5	0.3	54.36
36	0.5	0.9	0.155	0.5	0.9	0.3	48.31
37	0.7	0.5	0.3	0.5	0.5	0.3	50.8
38	0.5	0.1	0.155	0.5	0.9	0.01	30.58
39	0.3	0.1	0.155	0.3	0.5	0.155	32.25
40	0.5	0.5	0.01	0.3	0.5	0.3	34.89
41	0.3	0.9	0.155	0.7	0.5	0.155	47.25
42	0.5	0.5	0.155	0.5	0.5	0.155	47.43
43	0.5	0.5	0.155	0.5	0.5	0.155	49.29
44	0.5	0.5	0.01	0.7	0.5	0.01	37.13
45	0.3	0.1	0.155	0.7	0.5	0.155	33.52
46	0.5	0.1	0.01	0.5	0.9	0.155	29.44
47	0.3	0.5	0.01	0.5	0.5	0.3	34.2
48	0.5	0.5	0.01	0.7	0.5	0.3	35.61
49	0.5	0.5	0.155	0.5	0.5	0.155	47.83
50	0.3	0.5	0.155	0.3	0.9	0.155	52.14
51	0.5	0.1	0.3	0.5	0.1	0.155	33.72
52	0.5	0.9	0.155	0.5	0.1	0.3	46.67
53	0.5	0.5	0.3	0.7	0.5	0.3	54.5
54	0.7	0.5	0.01	0.5	0.5	0.01	41.46

TABLE A2 | Parameter combinations and numerical testing results for sample mass and disturbance rate.

No.	D (mm)	H (mm)	μ (-)	m_s (g)	I (%)
1	18	1	0.1	21.2535	12.5552
2	22	8.5	0.1	23.515	17.0102
3	18	16	0.1	18.1441	13.0667
4	18	8.5	0.5	6.93258	10.8244
5	18	8.5	0.5	7.59834	11.8402
6	22	8.5	0.9	6.11929	12.1322
7	22	16	0.5	9.07044	15.7702
8	22	1	0.5	10.3849	15.0753
9	14	1	0.5	4.99938	8.75214
10	18	8.5	0.5	7.14248	11.2552
11	14	8.5	0.1	14.3393	8.06589
12	18	8.5	0.5	8.12344	12.9495
13	14	8.5	0.9	2.09454	8.0965
14	18	1	0.9	8.19631	11.6723
15	14	16	0.5	4.38411	8.41759
16	18	16	0.9	5.52052	11.7796
17	18	8.5	0.5	7.94951	12.3429

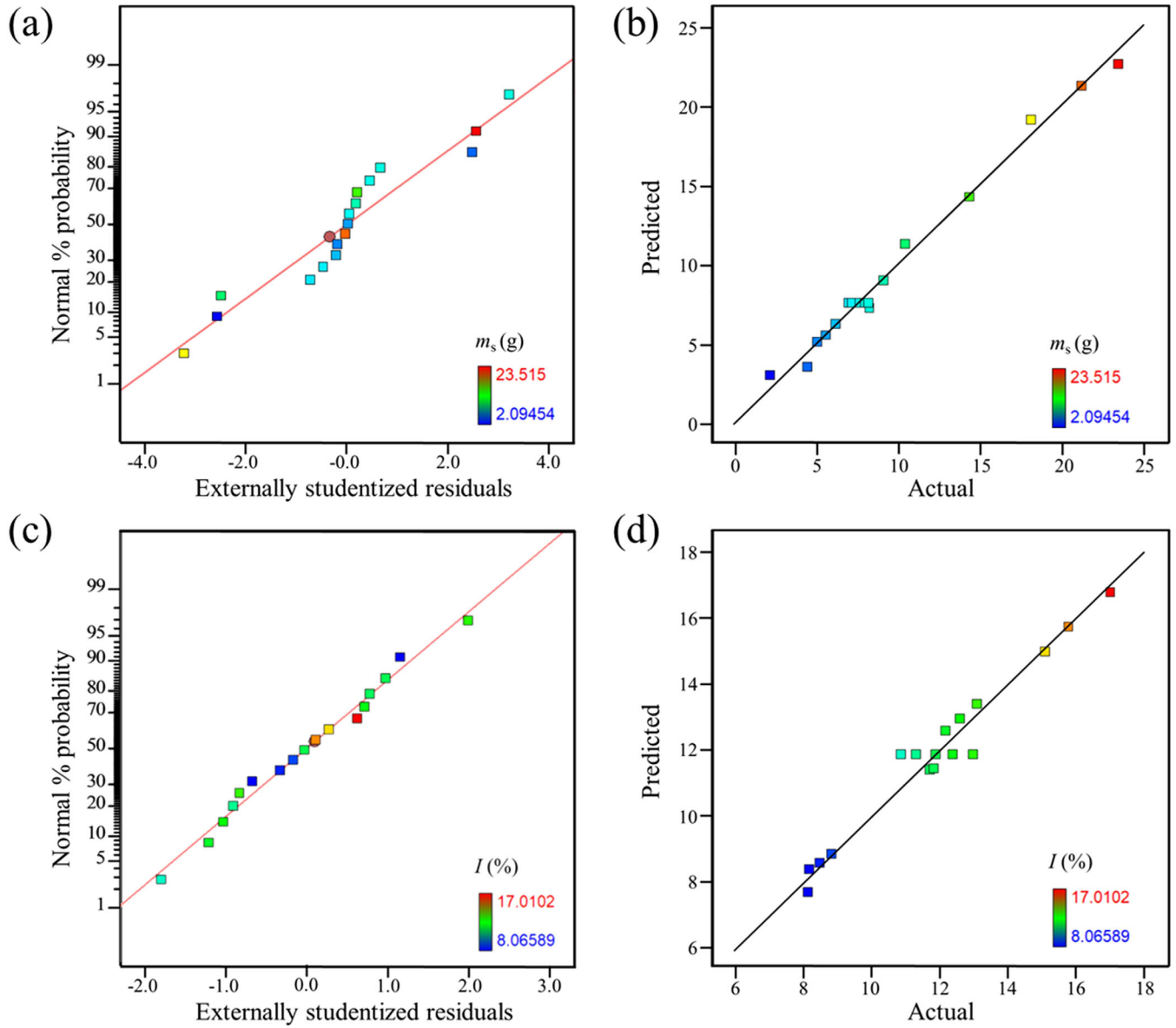


FIGURE A1 | Normal probability plots of the externally studentized residuals for RSD, where: (a) m_s ; (c) I ; correlations of results between numerical simulations and predictions, where: (b) m_s ; (d) I .